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This document was developed by a team led by the Secretary of the Environment, Carolina Urrutia Vásquez and coordinated by Sarah Arboleda Osorio. The core working group was composed of Ana Derly Pulido, Ricardo Delgado, Alejandra Cifuentes, Juan David Merlo, Óscar Santiago Vargas, Daniela Borbón, Edwin Lugo, Rodrigo González, Angie Lorena Téllez and Carolina Moscoso.

The extended working group of the District Environment Secretariat composed of Rafael Chaparro, Karen Blanco, Carlos Amaris, Liliana Castro Rodríguez, Germán Tovar, Eaking Ballesteros, Sandra Patricia Montoya Villarreal, María del Carmen Pérez, Carolina Torres, Mary Teresa Lizarazo, Hernán Gonzalo Cadena, José Fernando Cuello, Martha Molina, Adela Delgado, Lilian Bernal, Javier Cifuentes Álvarez, Édgar Alberto Rojas, Viviana Vanegas Vásquez, Nelson Betancourt, Héctor Julio Valbuena, Jairo Andrés Sánchez, Jorge Enrique Hernández, Juan Gabriel Alvarado, María del Pilar Rodríguez, Tatiana María De la Roche Todaro, Jaime Barrera Neira, Daniel Melquisedec Alayón, Jesús Alberto Martínez, David Sabogal Giraldo, Adriana Vega Romero, Daniel López, Carolina Amado, Javier David Sosa, Judith García, Nicolás Murillo, Julián Nicolás Grajales, Luz Stella Rey, Luz Helena Guzmán, Diana Carolina Mora Forero, Luis Felipe Jaramillo Giraldo, Deysi Milena Medina, Silvia Carolina Sánchez, Carolina Mariño Mogollón, María Alejandra Niño, Laura Cristina Castellanos, Pilar Viviana Díaz Gómez, Abdel Majid Assiz, Christian Valenzuela, Oliver Rivera, Károl Calceto, Natalia Sánchez, Diana Marcela Reyes, Luis Miguel Acosta and David Gerardo Calvo.

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Letter from Claudia Nayibe López
Mayor of Bogotá

Dear reader:

Climate change is no longer a matter of the future and, like most cities in the world, Bogotá has been affected by atypical climate events that confirm the urgency of achieving a commitment from all its citizens in the face of the climate crisis. The role of cities is clear: at a global level, they demand 80% of the world's energy, generate 70% of greenhouse gas emissions - GHG, and 50% of waste. Furthermore, the collateral effects on the urban population are systemic: diseases associated with air quality disproportionately affect the most vulnerable, and extreme weather events exacerbate poverty conditions which, in the case of a city like Bogotá, are the vast majority of urban inhabitants.

This administration is aware of the impact and urgency to vigorously confront this problem. For this reason, since February 2020, we promoted the declaration of a joint climate crisis with the departments of Cundinamarca, Boyacá, Meta and Tolima. This declaration included ten decisive mandates to take action and seek transparency in the reporting of the objectives set.

This Climate Action Plan articulates the roadmap for Bogotá to achieve the ambitious climate change mitigation and adaptation agenda that we have proposed. The evidence-based and carefully constructed actions that we propose to the city should lead us to a 15% reduction in GHG emissions by 2024, and 50% by 2030, thus making the city carbon neutral in 2050, as It is established by the District Development Plan: “A new social and environmental contract for the Bogotá of the 21st century.”

Changes in citizen habits are powerful tools to combat climate change from cities, and our Development Plan is based on that vision. The accumulated impact of the best decisions of citizens in terms of mobility, consumption and waste disposal,

added to the strong commitment of companies and public entities, will result in the climate action that is required so that it is not dramatically affected. human life on the planet.

Each of the sectors has a key role in achieving a responsible and resilient Bogotá. The update of the city's Greenhouse Gas Inventory confirms that household energy consumption, public and private transportation, and waste disposal are the three sectors that emit the most greenhouse gases and, therefore, , the mobility and habitat sectors have the best opportunity to contribute, from their policies and the way in which they are implemented, to the fulfillment of the climate action agenda of the country's capital.

Unlike at the national level, in Bogotá the change in land use is not one of the main sources of greenhouse gas (GHG) emissions. There is no doubt, however, that the conservation of the city's ecosystems is essential for mitigation and adaptation to climate change.

We have proposed, under criteria of climate and environmental justice, to declare corridors of protected areas in the south of the city that compensate for the historical deficit that the southwest has suffered in terms of green public spaces. While these new protection measures are consolidated, we have the obligation to fulfill our duties in the protection of the Cerros Orientales, our closest extensive forest. Its regeneration will be an important carbon sink; wetlands will fulfill their function of water regulation in the face of a changing climate; The conservation of the Sumapaz moor will be key to ensuring the water supply of millions of inhabitants, and the consolidated Thomas van der Hammen reserve, together with improved management of the Bogotá River that includes its decontamination, will be the flagship of citizen and institutional change.

This Climate Action Plan is a document that maps out our future. It is the way in which we plan to mitigate and adapt to the most important challenge of the 21st century, where institutions, the private sector and citizens have fundamental roles to ensure a prosperous future for the following generations. But, in addition to a roadmap, this document is an invitation for all of us to also assume, through our decisions and plans, a personal commitment to climate action.

CLAUDIA NAYIBE LÓPEZ HERNÁNDEZ
Mayor of Bogotá

Letter from Carolina Urrutia Vásquez
Secretary of the Environment of Bogotá

In Bogotá we have made the determination to give the priority it deserves to the climate crisis, based on what was established in the 2015 Paris Agreement, which contains the key commitments to avoid a 2 °C increase in global temperature with respect to preindustrial levels. According to the Intergovernmental Panel on Climate Change – IPCC, this limit must be raised to 1.5 °C, since exceeding it would cause catastrophic and irreversible climate events in the coming years. Without a doubt, Bogotá has assumed as its own the necessary measures to begin an urgent transition towards a more sustainable city model in harmony with nature.

The District Development Plan 2020-2024 "A new social and environmental contract for the Bogotá of the 21st century" lays the foundations to achieve significant progress in compliance with the Paris Agreement and the Sustainable Development Goals. Likewise, it seeks to ensure that those people born in 2020 can fully enjoy the conditions of this new social and environmental contract when they reach the age of majority and the 500 years of Bogotá are celebrated in 2038.

We are deeply convinced that climate goals are set by science, and governments must determine the routes that take us to that goal. Therefore, our goal of reducing emissions by 2030 is 50%, and our commitment is to lead the city to have net zero emissions in 2050, as established by scientists. To achieve these ambitious goals, my

administration committed to reducing greenhouse gas (GHG) emissions by 15% by the end of the four-year period, that is, by 2024.

The Administration of Mayor Claudia López has initiated processes that must be consolidated for many years to green Bogotá and mitigate and adapt to the climate crisis. Through the formulation of the Territorial Planning Plan (POT), which will be presented in 2021, we seek to create agreements and commitments with surrounding municipalities around the protection of our Main Ecological Structure, which includes the designations of protected areas, urban parks , ecological corridors and the special zone of the Bogotá River.

In addition, the POT seeks to ensure that the travel time of people in the city, whether from home to work or to any destination they are going to, does not exceed 30 minutes. In this way, we will reduce GHG emissions associated with mobility in Bogotá, where, currently, more than two hours are lost daily traveling. This is raised in the context of an economic recovery after the COVID-19 pandemic, which must be sustainable and in line with global commitments to mitigate and adapt to climate change.

We have also made the decision to take on leadership roles at the regional level. As vice president of the C40 steering committee, Bogotá will motivate more cities in Latin America to commit to effective climate action so they can jointly move towards a new, more environmentally smart normal.

This document offers an elaborate diagnosis of Bogotá's progress in climate matters and the challenges that we still must overcome. In addition, it proposes a series of commitments and a roadmap with clearly outlined dates that will allow us to position ourselves as a leading and ambitious city that manages to mitigate and adapt to climate change. However, this is not a finished document. The Plan must continue to grow and be constantly updated; Its compliance, the evolution of technologies for the energy transition and the growing interest and contribution of citizens and the private sector must ensure that the level of ambition and financing alternatives are increasingly greater and more abundant.

The periods of each administration are short compared to the time required for actions to stop the inertia of the past and modify the trends that have caused the climate crisis. We recognize the responsibility of proposing to the city, its organizations, companies and individuals, plans, programs, platforms and sufficient evidence so that a will for permanent evolution and greater ambition can be built on them.

In that sense, we welcome all approaches to improving and building together not only a plan, but the effective climate action that this city, our country and the planet need.

Carolina Urrutia Vásquez
Secretary of the Environment of Bogotá

Bogotá Climate Action Plan
2020 – 2050

Climate change is the greatest challenge facing humanity: it puts people's lives, health and emergency care systems, the economy, ecosystems and species, infrastructure and other interconnected and interdependent variables at risk. This challenge urgently demands the reduction of greenhouse gas (GHG) emissions, the strengthening of our resilience through adaptation to a changing climate and the ability to manage innovative and fair solutions for all citizens.

With more than seven million inhabitants and an emissions inventory of 11,421,724 tons of CO_2 equivalent in 2017, Bogotá has been strongly affected by the impacts of climate change in the last decade. During recent years, people in the city have suffered more intensely from the rainy and dry seasons, as well as their corresponding increase in floods; mass removal events, and the threat of energy rationing as a result of water shortages in droughts.

In the same way, the impact on the ecosystems of this biodiverse city has begun to be seen, bordered by the moors of Chingaza, Sumapaz and Cruz Verde, embraced by the Cerros Orientales and adorned by a complex of more than fifteen wetlands. Our ecosystems are the first responders to the crisis and, for this reason, their care and conservation is a priority.

Bogotá recognizes the urgency of confronting climate change in a comprehensive

manner and based on collective effort. Consequently, the district Administration proposes to the city the following Climate Action Plan (CAP) as an instrument of strategic direction for the Capital District.

This document identifies and prioritizes tools and mechanisms for mitigation and adaptation to climate change from the specific local context of the city. Additionally, it builds on past efforts, such as the 2015 District Climate Change and Risk Management Plan and the 2018 District Climate Change and Disaster Risk Management Plan, to propose a specific approach to climate change to Bogotá from a transversal perspective that grants responsibilities to all sectors, companies and citizens.

This plan is designed as a living organism that must be updated periodically, in periods of no more than five years, and based on two parallel processes. The first corresponds to the progress in the implementation of the included measures and the emergence of new alternatives and technological advances, which will present more and better options for adaptation and mitigation. The second responds to the way in which, without a doubt, the interest and participation of all sectors of citizens in climate action will be incrementally amplified.

As a priority, beyond presenting the city with a roadmap, this PAC is an open invitation to citizens, companies and organizations to propose, implement and ensure that Bogotá becomes an increasingly sustainable and sustainable city. resilient.

2

Transverse characteristics

The 2020-2050 Climate Action Plan has three transversal characteristics:

1. **Effective mitigation and adaptation actions to climate change**

This plan proposes the necessary actions so that by 2050 Bogotá is carbon neutral and resilient to climate events. The above, in accordance with the commitments and goals defined in the 2015 Paris Agreement.

These measures seek to identify and take advantage of synergies and interdependencies so that each one contributes to the strengthening of other actions. This enhances your efficiency and reduces investment risk.

2. **Actions based on evidence, technically viable, and with monitoring mechanisms**

The actions and goals outlined in this Climate Action Plan arise from a rigorous technical process that guarantees their relevance, feasibility and final impact.

All the actions described were designed and prioritized by an interdisciplinary group of experts from the different entities of the district Administration. These also respond to diagnoses and models of greenhouse gas emissions and climate risk, prepared by the District Environment Secretariat and the District Institute for Risk Management and Climate Change (Idiger) and presented in this document.

It should be noted that compliance and impact of the actions will be reported

through specific monitoring, evaluation and results reporting mechanisms.

3. Participatory elaboration and inclusive implementation

This Climate Action Plan emerges from a participatory and iterative process with citizens and key sectors of the city.

The document emphasizes the leadership role that the District must assume to mobilize the private sector, academia and civil society towards proactive climate action, which transcends regulations and cooperates with other cities, organizations and entities at the regional, national and international.

On the other hand, the proposed actions are inclusive and equitable in their implementation, prioritizing the most vulnerable population in the city and incorporating a differential territorial, socioeconomic and gender approach.

Methodology

The comprehensive climate bets of this PAC were developed in accordance with the methodology of the C40 Climate Leadership Group within a period of 20 weeks, between the months of June and December 2020.

This process included the following activities:

- **Build on what is built** through the review of current and past, national and district policy and planning instruments and documents to collect the actions previously defined by the city and align them with what is established by the country.

- **Strategic survey** to analyze the city's development priorities and objectives and ensure that measures provide benefits to multiple stakeholders.

- **Vision, commitment and involvement workshops** with key actors from the public and private sectors, academia and civil society grouped in the Intersectoral Commission for Risk Management and Climate Change (CIGRCC) and the Advisory Council for Risk Management and Climate Change (CCGRCC).

  During these meetings, held between June and October 2020, it was agreed on the need to update the old District Climate Change Risk and Disaster Management Plan (PDGRCC) through this Climate Action Plan. Likewise, priority measures began to be identified for the energy, transport, waste, services and climate risk sectors.

- **Production of a document on climate risk** based on a review of information present in the District and dialogues with Idiger experts.

- **Construction of a list of mitigation and adaptation actions** in accordance with the objectives of the Paris Agreement and the city's goals included in different regulatory and policy instruments.

- **Virtual workshops for selecting criteria for prioritization and rating of actions,** with emphasis on its direct impact on climate change mitigation and adaptation, as well as co-benefit and feasibility criteria, based on the ASAP Tool methodology.

- **Virtual workshop to identify priority mitigation and adaptation actions** based on previously defined criteria.

- **Virtual workshops to identify synergies with other social sectors** to observe the good practices that are being developed from the private sector, civil society, academia and localities in terms of climate action. The priorities of these groups were also recognized, as well as the needs for support and leadership from the District.

Likewise, socialization activities of the Climate Action Plan began with leaders and representatives of the localities within the framework of the Local Environmental Commissions (CAL).*)*. As of December 2020, four meetings of this type had been held with the CALs of La Candelaria, Puente Aranda, Suba and Usme. It is expected to continue deepening these actions until the delivery of the final version of the PAC in June 2021.

4

Document structure

This document is structured into five central chapters:

I. **Geographic and socioeconomic profile of Bogotá.**
In this part, geographical characteristics are provided and socioeconomic trends are discussed. The context of the COVI-19 pandemic is also included.

II. **Diagnosis of greenhouse gas emissions and climate risk analysis.**
The results of the Greenhouse Gas Emissions Inventory, the emissions trajectory, the carbon budget and the climate risk and impact assessments are analyzed.

III. **Mitigation and adaptation goals and objectives 2020-2050.**
The main milestones for mitigation and adaptation to climate change are defined in the period 2020-2050, in line with the commitments of the Paris Agreement and the post-coronavirus economic recovery agenda.

IV. **Responsibilities, synergies and participation**.
This chapter establishes a mapping of the main district entities responsible for the implementation of the Climate Action Plan and discusses the District's leadership priorities at the regional, national and international level, as well as the forms of mobilization of other social sectors and the citizenship around the

outlined actions.

IN. **Climate actions.**
The specific actions and sub-actions for mitigation and adaptation to climate change, as well as the transversal component, adopted by the city are presented and their role in compliance with the 2020-2050 roadmap is discussed.

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7

SCHURCH AND ACRONYMS

ANDI National Association of Businessmen of Colombia

ANLA National Environmental License Authority

ASOCAPITALES Colombian Association of Capital Cities

C40 C40 Climate Leadership Group

CADE Specialized District Care Center

CAEM Business Environmental Corporation

CAL Local Environmental Commission – Local Environmental Commissions CAR

Regional Autonomous Corporations

CAR Regional Autonomous Corporation of Cundinamarca

CCB Chamber of Commerce of Bogotá

CCGRCC Advisory Council for Risk Management and Climate Change CDA Automotive

Diagnostic Center

CDPMIF District Commission for the Prevention and Mitigation of Forest Fires UCLG United

Cities and Local Governments

CIDEU Ibero-American Center for Strategic Urban Development CIGRCC Intersectoral

Commission for Risk Management and Climate Change UNFCCC United Nations

Framework Convention on Climate Change CPL Local Planning Council

CREG Energy and Gas Regulation Commission

CAV Popular Housing Fund

DADEP Administrative Department of the Public Space Ombudsman DCC Colombian

Civil Defense

DNP National Planning Department

DOTS Sustainable Transportation Oriented Development

EAAB Aqueduct and Sewerage Company of Bogotá

EEP Main Ecological Structure

EIC Heat Island Effect

EMB Bogotá Metro Company

ERC Risk Assessment in Climate Change Scenarios ESI-SARI Influenza-Like Disease - Severe Acute Respiratory Infection FITIC Fund for Innovation, Technology and Creative Industries FNCER Non-Conventional Sources of Renewable Energy FONDIGER District Fund for Risk Management and Climate Change GEB Bogotá Energy Group

GHG Greenhouse Gases

ICLEI Local Governments for Sustainability

IDEA Corporate Environmental Performance Index

IDEAM Institute of Hydrology, Meteorology and Environmental Studies IDIGER District Institute of Risk Management and Climate Change IDPAC District Institute of Community Participation and Action IDRD District Institute of Recreation and Sports

IDU Institute of Urban Development

IGAC Agustín Codazzi Geographic Institute

INGEI National Greenhouse Gas Inventory IPCC Intergovernmental Panel on Climate Change IPES Institute for Social Economy

JAC Community Action Boards

JBB Bogotá Botanical Garden José Celestino Mutis

MADS Ministry of Environment and Sustainable Development

MVCT Ministry of Housing, City and Territory

NDC Nationally Determined Contributions

NRCOA Regional Node of the Andean East Center

OAB Environmental Observatory of Bogotá

SDG Sustainable Development Goals

PAC Climate Action Plan

PAL Local Environmental Plans

PDGRCC District Plan for Climate Change Risk and Disaster Management PNN National Natural Park

UNDP United Nations Development Program POMCA Planning and Management Plans for Hydrographic Basins POT Territorial Planning Plan

PRAES School Environmental Projects

PRAU University Environmental Projects

PREAD District Environmental Excellence Program

PRICC Comprehensive Regional Plan for Climate Change of Bogotá-Cundinamarca

Pro-RedES Corporate Responsibility and Sustainability Projects PROCEDA Community Processes of Environmental Education

RAPE Administrative and Special Planning Region

RMCAB Air Quality Monitoring Network

SAB Alert System of Bogotá

SDA District Environment Secretariat

SDCRD District Secretariat of Culture, Recreation and Sports SDDE District Secretariat of Economic Development

SDGR-CC District Risk and Climate Change Management System SDHT District Habitat Secretariat

SDM District Secretary of Mobility

SDP District Planning Secretariat

SED District Secretary of Education

SEGAE Subdirectorate of Ecourbanism and Business Environmental Management

SGSSS General System of Social Security and Health

SIGAU Information System for the Management of Urban Trees in Bogotá SISCLIMA National

Climate Change System

SITP Integrated Public Transport System

SUDS Sustainable Urban Drainage Systems

TGI Transportadora de Gas Internacional SA ESP

UAECD Special Administrative Unit of District Cadastre

UAECOB Special Administrative Unit of the Official Fire Corps of Bogotá

UAESP Special Administrative Unit of Public Services UCCI Union of

Ibero-American Capital Cities ICU Intensive Care Unit

UDFJC Francisco José de Caldas District University UMV Road

Maintenance Unit

UPME Mining-Energy Planning Unit

UPR Rural Planning Unit

UPZ Zonal Planning Unit

11

GLOSARIO

Adaptation to change climate

Adjustment process to the present and expected effects of climate change. In social decision-making areas it corresponds to the adjustment that seeks to mitigate the harmful effects and take advantage of the current or future beneficial opportunities of the climate and its effects.

Threat

In socioecosystems, the process of adjustment of biodiversity to the current climate and its effects can be intervened by society with the purpose of facilitating the adjustment to the expected climate (Law 1931 of 2018).

origin, caused or induced by human action accidentally, occurs with sufficient severity to cause loss of life, injuries or other impacts on health, as well as damage and detriment to property, infrastructure, livelihoods, service provision and environmental resources (Law 1523 of 2012).

Latent danger that a physical event of natural

Anthropogenic Resulting from the activity of human beings or a product of this.

Variation in the state of the climate, identifiable, for example, by statistical tests, in changes in the mean value or variability of its properties, which persists over long periods, usually decades or longer. Climate change may be due to processes

Climate change
natural internal forces or external forcings such as modulations of solar cycles, volcanic eruptions or persistent anthropogenic changes in the composition of the atmosphere due to increases in greenhouse gas concentrations or land use.

Climate change could modify characteristics of extreme meteorological and hydroclimatic phenomena in their average frequency and intensity, which will gradually be expressed in their spatial behavior and annual cycle (Law 1931 of 2018).

greenhouse

Co-benefits

Low carbon and climate resilient development

Scenery
climate
Positive effects that a policy or measure aimed at one objective could have on others, regardless of the net impact on general social

Effect

well-being. Co-benefits are often subject to uncertainty and depend, among other factors, on local circumstances and implementation practices. These are often called secondary benefits (IPCC, 2014). (Law 1931 of 2018).

Development that generates a minimum of greenhouse gases and manages appropriately address climate-related risks to reduce vulnerability, while making the most of opportunities to development that climate change generates.

Natural phenomenon by which the earth retains a part of the solar energy in order to maintain a temperature that enables natural development of the living beings that inhabit it.

Plausible and sometimes simplified representation of the future climate, which is based on an internally coherent set of climatological relationships, explicitly defined to investigate the possible consequences of anthropogenic climate change, which can be input as input into impact models.

Climate projections are often used as a starting point to define climate scenarios, although these usually require additional information such as the current observed climate (IPCC, 2018).

Exhibition Presence of people, livelihoods, environmental services and economic and social resources, cultural assets and infrastructure that, due to their location, may be affected by the manifestation of a threat (Law 1523 of 2012).

Frequency Number of events per defined unit of time (ISO/IEC, 2009).

climate

gases
effect
winter (GEI)

Management
risks

Change management

monitoring and evaluation of policies and permanent actions aimed at increasing knowledge of risk and promoting greater awareness of it, preventing or avoiding its generation, reducing or controlling it when it already exists and offering preparation for manage disaster situations, as well as subsequent recovery, understood as rehabilitation and reconstruction. These actions have the explicit purpose of contributing to the security, well-being and quality of life of people and sustainable development (Law 1523 of 2012).

Impacts
(consequences and results)

Gaseous components of the atmosphere, of natural or anthropogenic origin, that absorb and emit solar energy reflected by the earth's surface, the atmosphere and clouds. The main greenhouse gases are carbon dioxide (CO2), nitrous oxide (N20), methane (CH4), hydrofluorocarbons (HFC), perfluorocarbons (PFC) and sulfur hexafluoride (SF6).

In this report, the term “impacts” is primarily used to describe the effects of extreme weather and climate events and climate change on natural and human systems.

Impacts generally refer to effects on lives, livelihoods, health, ecosystems, economies, societies, cultures, services and infrastructure, due to the interaction of dangerous climate changes or phenomena that occur in a specific period of time and the vulnerability of societies or systems exposed to them. These are also called “consequences” and “results.”

Coordinated process of design, implementation and evaluation of GHG mitigation actions and adaptation to climate change, aimed at reducing the vulnerability of the population, infrastructure and ecosystems to the effects of climate change. It includes actions aimed at allowing and taking advantage of the opportunities it generates.

Social process of planning, execution,

greenhouse gas emissions in the atmosphere through the limitation or reduction of their emission sources and the increase or improvement of their sinks and reserves.

Climate change mitigation includes policies, programs, projects, incentives or disincentives and activities related to the Colombian Low Carbon Development Strategy and the National REDD+ Strategy (ENREDO+).

Greenhouse gas mitigation

Measure of the possibility of an event occurring. It can be defined, measured or determined and is represented qualitatively or quantitatively in terms of probability or frequency (ISO/IEC, 2009).

Probability

Simulated response of the climate system to various scenarios of future emissions or concentrations of greenhouse gases and aerosols, often based on simulations using climate models.

Projection

Climate projections differ from climate predictions due to their dependence on the emission, concentration or radiative forcing scenario used, based on relative assumptions such as, for example, a socioeconomic and technological future that may or may not materialize (IPCC, 2014).

climate

The impacts of climate change on geophysical systems, including floods, droughts and sea level rise, are a subset of so-called “physical impacts”. (IPCC, 2014).

Management that seeks to reduce the levels of

Resilience or ability to adapt
The ability of social, economic and environmental systems to cope with a dangerous event, trend or disturbance to respond or reorganize in a way that maintains its essential function, identity and structure, and, at the same time, retains the capacity for adaptation, learning and transformation (Law 1931 of 2018).

Risk Interaction of exposure and vulnerability in relation to a specific hazard (IPCC, 2014).

Risk associated with change climate Potential consequences in which something of value is in danger with a uncertain outcome, recognizing the diversity of values. Risks result from the interaction of vulnerability, exposure and threat. In this Law, the term risk is used mainly in reference to the risks associated with the impacts of climate change (Law 1931 of 2018).

Climate sensitivity Degree to which a system is affected, either adversely or favorably, by events associated with climate change. It allows identifying population groups that are more vulnerable to the threats associated with climate change and elements that make the population more vulnerable. Economic and physical related to infrastructure (IPCC, 2014).

Vulnerability Susceptibility or physical, economic, social, environmental or institutional fragility to suffer adverse effects in the event that a physical event associated with a hydroclimatological phenomenon occurs. It corresponds to the predisposition of human beings and their livelihoods to suffer loss or damage, as well as the deterioration of ecosystems, biodiversity, ecosystem services, water resources and the physical, social, economic and support systems that can be affected (Law 1931 of 2018).

1. PGEOGRAPHICAL AND SOCIOECONOMIC PROFILE OF BOGOTÁ

1.1. Geographic context

Bogotá is the capital of the Republic of Colombia and the department of Cundinamarca. It is located to the east of the Bogotá Savannah at a height of 2,630 meters above sea level, located on the Cundiboyacense plateau, belonging to the mountain formation of the Eastern Andes mountain range. To the east, it has its natural border in the Cerros Orientales, which extend in a south-north direction, and to the south, its rural area includes a part of the Sumapaz moor, considered the largest in the world, located about an hour away. of the city. The Bogotá River acts as the city's northwestern border.

The capital is organized into twenty locations: Usaquén, Chapinero, Santa Fe, San Cristóbal, Usme, Tunjuelito, Bosa, Kennedy, Fontibón, Engativá, Suba, Barrios Unidos, Teusaquillo, Los Mártires, Antonio Nariño, Puente Aranda, La Candelaria, Rafael Uribe Uribe, Ciudad Bolívar and Sumapaz. The total territorial extension is 163,635 hectares, of which 37,972 (23.2%) correspond to the urban area, 122,687 (75%) to the rural area and 2,974 (1.8%) to expansion land.

The rural territory is distributed in nine (9) of the twenty localities: Sumapaz, Usme, Ciudad Bolívar, Usaquén, Santa Fe, San Cristóbal, Chapinero, Suba and Bosa, which are home to subsistence farmers, small agricultural entrepreneurs, and Muisca indigenous communities. and landowners and flower manufacturers. 70% of the land in rural Bogotá is made up of paramo ecosystem, 9.1% of high Andean forest and scrubland, 1.6% of forest plantations, 15.5% of pastures and 2.9%. for crops such as potatoes and vegetables.

Through Legislative Act 002 of 2020, the Congress of the Republic modified article 325 of the Political Constitution of Colombia to create “the Bogotá–Cundinamarca Metropolitan Region as an administrative entity of regional associativity with a special regime, with the objective of guaranteeing the execution of sustainable development plans and programs and the timely and efficient provision of the services under their responsibility.” In this sense, and in line with the Legislative Act, the Mayor's Office of Bogotá and the Government of Cundinamarca will submit to a vote of the Council of Bogotá and the Departmental Assembly the entry into the Metropolitan Region, and an organic law will define its operation.

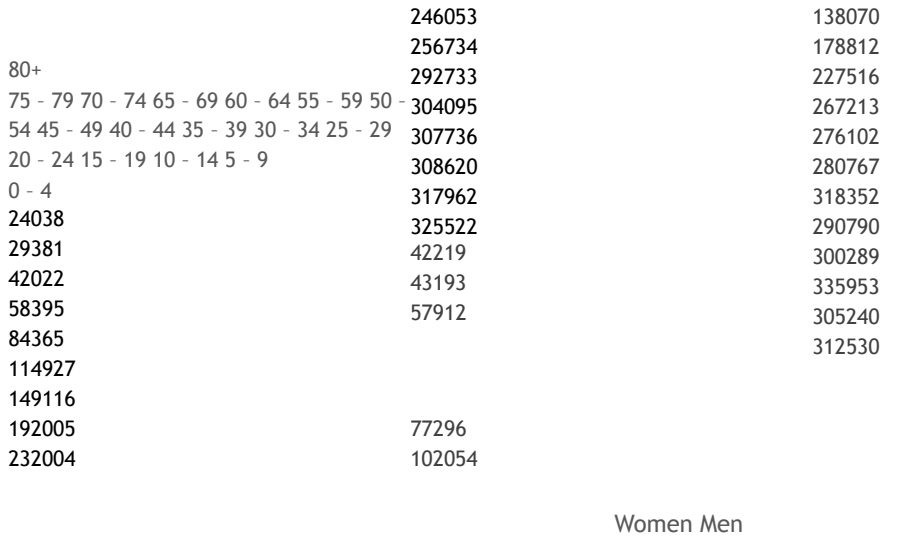
1.2. Socioeconomic context

1.2.1. Demographics and quality of life

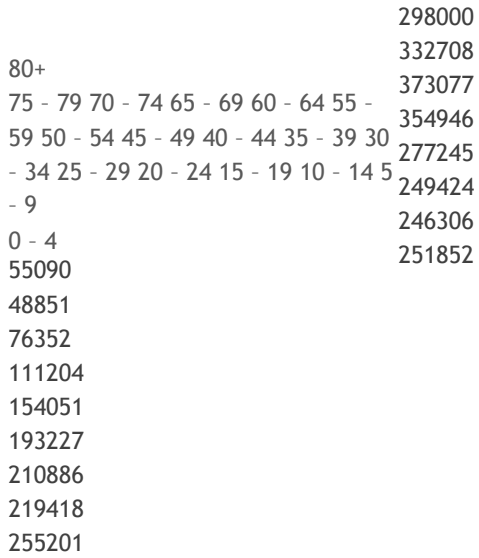
Bogotá has 7,412,566 inhabitants, according to the 2018 National Population and Housing Census. According to its projections, the city will have a total of 8,434,700 in 2030 and 9,164,445 in 2050. As shown in Graph 1 and the

Graph 2, the population pyramid of the District has shifted in the last 15 years, which demonstrates a general aging of the population.

Graph 1 Population pyramid of Bogotá, D.C. (2005)



Graph 2 Population pyramid of Bogotá, D.C. (2020)



87018
70936
106789
149260
198658
241982
257211
257705
284724
315704
340455
376169
356644
273634
241276
236517
241435

Women Men

Source: District Health Secretariat. Bogotá Health Observatory-SaluData 2020

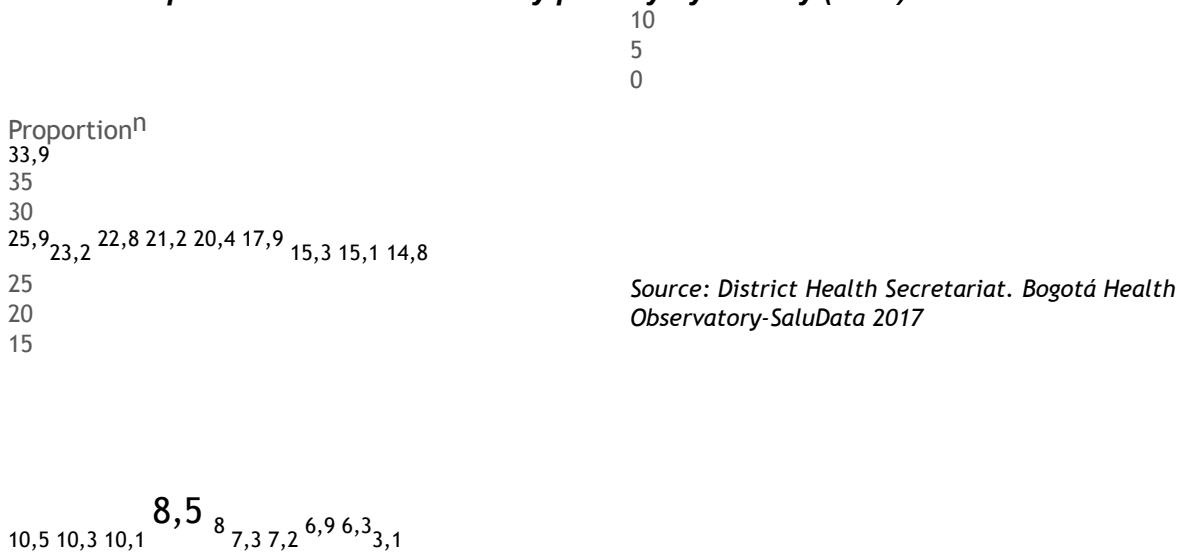
The city has taken important steps in improving the living conditions of its inhabitants. As shown in Graph 3, which indicates that the incidence of monetary poverty was reduced by almost a third between 2002 and 2018.

Graph 3 Incidence of extreme monetary poverty

Source: District Health Secretariat. Bogotá Health Observatory-SaluData 2018

However, there are still important challenges when it comes to distributing these benefits throughout the district territory. The proportion of people with unsatisfied basic needs, which corresponds to 3.34% in urban areas, rises to 11.03% in rural areas according to the 2018 National Population and Housing Census. Likewise, deep economic inequalities persist in the interior of the urban area. Graph 4 shows the incidence figures of monetary poverty, differentiated by locality, as of 2017. Large economic gaps are observed between the localities of Usme, Ciudad Bolívar, Santa Fe and Tunjuelo, and those of Barrios Unidos, Usaquén, Chapinero and Teusaquillo . These shortcomings in equitable distribution show the need to achieve a new social contract in Bogotá.

Graph 4 Incidence of monetary poverty by locality (2017)



Source: District Health Secretariat. Bogotá Health Observatory-SaluData 2017

1.2.2. Industry and employment

In the Bogotá-Cundinamarca region, 28% of national production is generated and around 35% of the country's companies are concentrated. According to data published by the Bogotá Chamber of Commerce (CCB), in 2018 there was an increase of 4.9% in the number of active companies and establishments in the Commercial Registry compared to the previous year, going from 728,784 productive units to 764,639.

Of the total, 487,148 are natural and legal persons, and 277,491 are commercial establishments. The services sector leads with 363,705 companies and commercial establishments (47.5%), followed by commerce with 259,195 (33.9%) and industry with 135,931 (17.7%). Suba is the town with the largest number of companies and commercial establishments, having 77,638, which represents 12.1%; Kennedy follows, with 65,442 (10.2%); Engativá, with 62,720 (8.2%); Usaquén, with 59,359 (7.8%); and Chapinero, with 54,230 (7.1%).

COVID-19 had a profound impact on the District's industry and employment. According to the National Covid-19 Economic Impact Survey, carried out by the Bogotá

Chamber of Commerce, as of April 2020, between 370,000 and 550,000 jobs had been lost, generating an unemployment rate of between 18 % and 22%.

Teleworking and the flexibility of working conditions has been the main strategy of employers in the face of the situation caused by the pandemic; However, 37% of companies temporarily ceased their activities and 15% requested bank loans at the end of April 2020. Likewise, the number of companies registered and renewed in Bogotá in the first quarters of 2019 and 2020 decreased radically: from 252,559 microenterprises and 26,388 small companies in 2019 increased to 145,784 microenterprises and 7,822 small companies in 2020.

1.2.3. Health

According to the National Quality of Life Survey, in 2019, 91.3% of respondents stated that they were affiliated with the General Social Security and Health System (SGSSS), that is, 1.5 points below the national average.

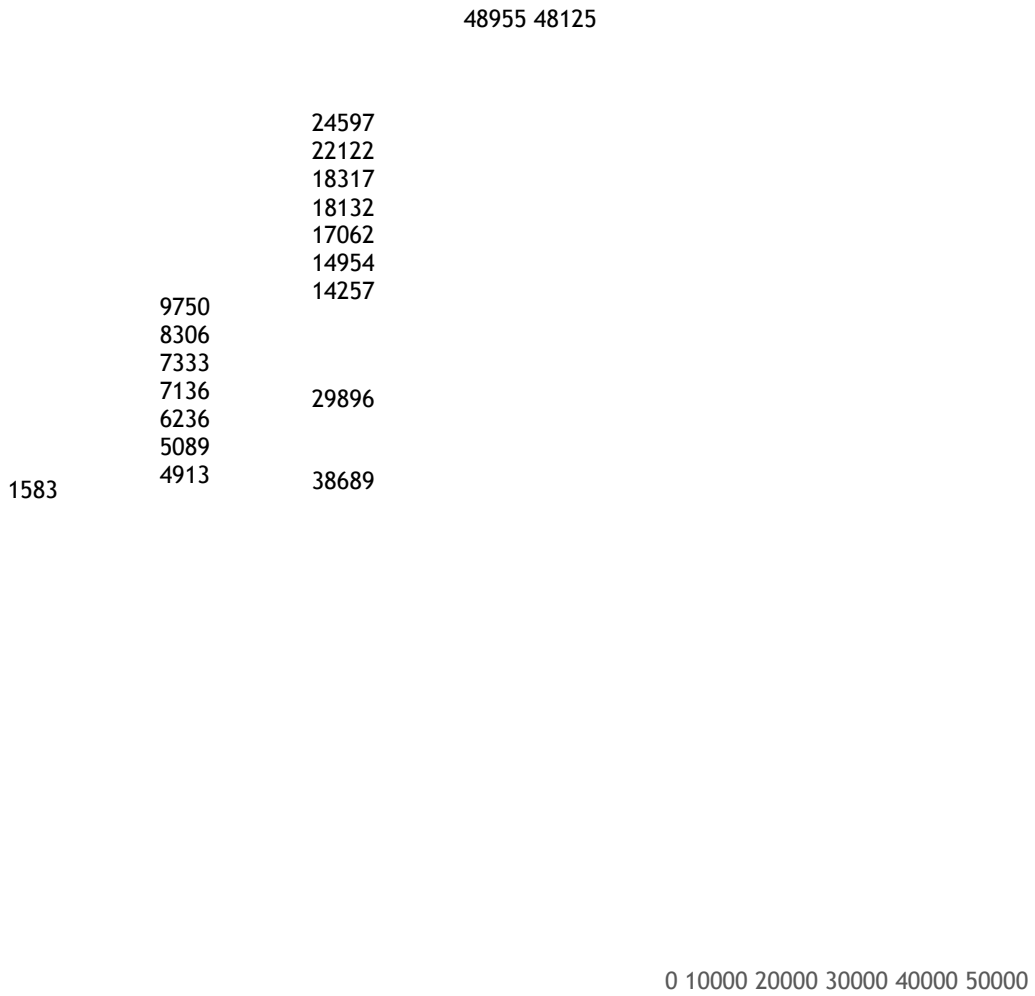
Weather patterns have a significant impact on health in Bogotá. Between 2009 and 2019, the highest numbers of care for acute respiratory illness and the mortality rate due to acute respiratory infection in children under five years of age occurred during the years 2011 and 2012 and coincided with the La Niña climate event, accompanied by heavy rainfall. Likewise, when observing the notifications of influenza-like illness – severe acute respiratory infection (SARI-SARI) between 2009 and 2019, it is evident that 60% of the cases occurred in the rainy season, with a valley in the 2015, which coincided with the lowest accumulated annual precipitation of the period and the El Niño climate phenomenon, as reported by the Bogotá Health Observatory SaluData.

28.5% of confirmed cases of COVID-19 in Colombia are concentrated in Bogotá. As of December 1, 2020, there were 377,229, of which 54.6% were located in the towns of Suba, Kennedy, Engativá, Bosa and Ciudad Bolívar, as evidenced in

Graph 5. By that date, 347,323 people had recovered, 5% were in mild condition, 0.2% in moderate and 0.2% in severe condition. The death rate was 2.3%. Thanks to the increase in Intensive Care Units (ICUs), between May and December 2020, their occupancy remained, to a greater extent, below 75%, except for June and August, when the threshold was exceeded. The peak ICU occupancy for said period was reached on July 13 (97.3%).

Graph 5 COVID-19 cases by locality as of December 1, 2020

	Cristobal	Antonio
Below	Fontibón	Nariño La
Engativá	use me	Candelaria
Ciudad	Chapinero	
Bolívar San	Teusaquillo	



Source: District Health Secretariat. Bogotá Health Observatory-SaluData 2020

1.2.4. Mobility

As of December 2019, the inhabitants of Bogotá made 4,556,702 daily trips by public transport, 1,986,760 by private vehicle and 880,367 by bicycle, according to the Mobility Survey. In lower strata, trips on foot predominate (between 22% and 32%), by TransMilenio (18% to 24%), motorcycle (5% to 8%) and bicycle (6% to 9%), and in upper strata The car (32% and 46%), taxi (9% - 12%) and informal transport (4% to 6%) prevail. The average travel time for unimodal trips, that is, in which only one type of transportation is used, is 57 minutes.

Within the framework of the Mobility Master Plan, the Integrated Public Transportation System (SITP) of Bogotá integrates a scheme of trunk routes - which correspond to the TransMilenio System -, zonal and feeder routes. At the end of 2018, the first TransMiCable line was inaugurated between El Tunal and Mirador del Paraíso, which shortened travel times during rush hour from one hour to 13 minutes. The system has 114.4 kilometers (km) of trunk infrastructure, 1889.6 km of zonal routes, 441 km of feeders and 3.3 km of cable.

Between 2011 and 2019, the number of daily bicycle trips doubled (from 441,135 to 880,367). The urban area has more than 500 km of cycle paths, that is,

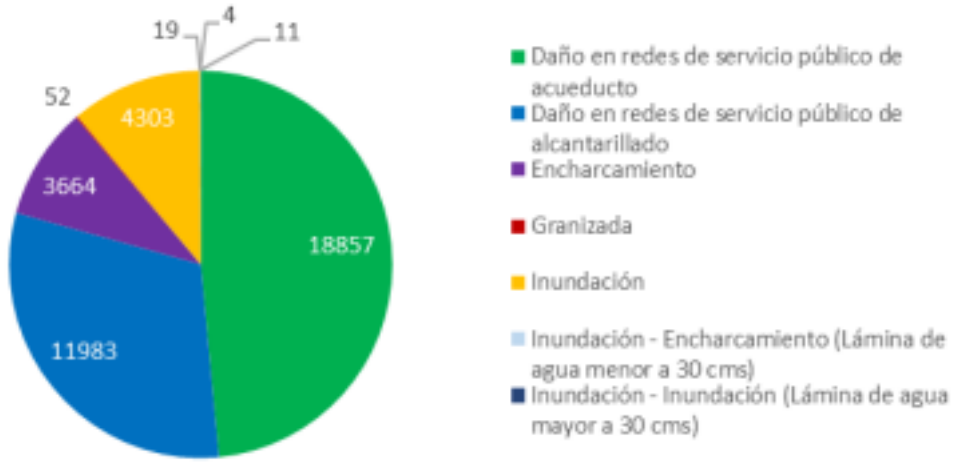
urban infrastructure of exclusive lanes for the circulation of bicycles and similar. In addition, on Sundays and holidays the District enables a 127.69 km cycle path circuit throughout the urban area to encourage physical activity as a family. He *Copenhagenize Index* 2019 placed Bogotá in 12th place in its global ranking "*The Most Bicycle-Friendly Cities of 2019*" to make important recognition of the favorable conditions faced by cyclists in the city.

1.2.5. Infrastructure

The infrastructure provision aspects change a lot between one sector and another. Access to road infrastructure is almost homogeneous in Bogotá, but this is not the case with cultural, educational or sports facilities. Although the city has carried out a decentralization strategy since the first years of the 21st century with systems such as the Specialized District Care Centers (CADE) and local management and coordination in health, education and social management, the mountain and Southeast and Southwest suffer from less accessibility. In the same way, this manifests itself in transportation networks, which constitute barriers to access to the city, work and urban services.

The main infrastructure vulnerabilities are associated with extreme rainfall phenomena which, according to the Information System for Risk and Climate Change Management (SIRE) of Bogotá, correspond to 80% of the events attended in the period 2006-2018. , as referenced in Graph 6. This situation has reduced over time and it is expected that, with the recognition of the need for renaturalization of previously waterproofed areas, as well as with the implementation of sustainable urban drainage systems (SUDS), continue along the same path.

Graph 6 Sources of climate vulnerability in infrastructure (2006-2018)



Source: Own elaboration with data from SIRE 2006-2018

1.3. Environmental aspects

1.3.1. Main Ecological Structure (PES)

The Main Ecological Structure (EEP), established by Law 388 of 1997, is a network of spaces and corridors that sustain and conduct biodiversity and essential ecological processes through the district territory, in their different forms and intensities of occupation, to provide it with environmental services aimed at its sustainable development.

According to article 16 of Decree 190 of 2004, "by means of which the provisions contained in District Decrees 619 of 2000 and 469 of 2003 are compiled", the EEP of the capital "is constituted by a network of environmental corridors located in jurisdiction of the Capital District and integrated into the Regional Ecological Structure", whose basic components are described below:

✔ *District Protected Area System*

It is the set of spaces with unique values for the natural heritage of the District, the region and the nation. All its elements are protective floor. This is classified like this:

- National and regional protected areas: at the national level there is the Sumapaz National Natural Park and the Bosque Oriental de Bogotá Protective Forest Reserve, and at the regional level there is the Northern Regional Forest Reserve Area.

- Protected areas of the district order: the District Fauna and Flora Sanctuaries, the District Forest Areas and the District Ecological Parks form part of this category.

✔ *Metropolitan and urban parks*

They group together those elements of public space intended for recreation. These fulfill the function of establishing the spatial connection between the elements of the Protected Area System to give continuity to the structure. Within the metropolitan and urban parks component there are the categories: 1) passive recreation (rivers and canals); and 2) active recreation.

✔ *Special Management Area of the Alluvial Valley of the Bogotá River*

Included in the system of District Protected Areas is the hydraulic circuit and the management and environmental preservation zone of the river, defined in order to coordinate the District's actions to enhance it as the main axis of articulation with the regional context.

✔ *Cerros Orientales - Eastern Forest Protective Forest Reserve of Bogotá*

These are part of the Protected Areas System of the Capital District and, for planning and investment purposes, they are subject to the regime of uses, management plans and specific regulations.

✔ *Ecological corridors*

They are linear green areas that follow the urban edges and main components of the water network and arterial road network as part of their environmental management with the objective of increasing the ecological connection between the other elements of the Main Ecological Structure: from the Eastern Hills to the Special Management Area of the Bogotá River and other rural and urban areas.

The elements and components of the Main Ecological Structure are oriented towards its integration with a Regional Main Ecological Structure, within the framework of the hydrographic subzone of the Bogotá River, belonging to the Magdalena-Cauca hydrographic area and the set of strategic ecosystems that compose it. For this reason, the Special Management Area of the Bogotá River and, in general, the System of Protected Areas act as integrating axes that guarantee ecosystem processes and the provision of environmental services in the region.

1.3.2. Hydrography

The Eastern Hills determine the water characteristics of the city and are a fundamental part of the formation of its hydrographic network, which is mainly made up of the basins of the Salitre or Juan Amarillo, Fucha and Tunjuelo rivers, which drain superficially into the riverbed. of the Bogotá River, in its middle basin, and, in turn, in a southerly direction towards the Magdalena River, in the municipality of Girardot (Tolima).

Below is a brief description of the mentioned basins: ● ***Salitre River Basin***

It is located in the central-northern sector of the Capital District and has a drainage area of 13,964 hectares. It has a main channel length of 19.76 km and its average slope is 3.32%. Its average height is 2,870 m above sea level. n. m., where the maximum elevation is around 3,200 m above sea level. n. m. and the minimum above 2,540 m above sea level. n. m. approximately.

The Delicias, La Vieja, La Chorrera, Catalonia, La Canada, Luce, Moraji, Chico, Los Cerros and Los Rosales streams, as well as the Jaboque, Santa María del Lago, Córdoba and Juan Amarillo or Tibabuyes wetlands[1]They are part of this basin.

● ***Fucha River Basin***

Due to its conformation and length, it is subdivided into three sectors: the upper basin, corresponding to the mountainous part; the middle basin, which begins at the river's entrance to the urban perimeter of Bogotá and extends to Boyacá Avenue; and the lower basin, between Boyacá Avenue and its mouth into the Bogotá River.

[1]Bogotá District Environment Secretariat (SDA), 2015. Description and context of the water basins of the capital district (Torca, Salitre, Fucha and Tunjuelo). Mayor's Office of Bogotá. Bogotá, D.C.

Thus, the Fucha River is one of the most important water bodies in the Bogotá Savannah with a length of 17.30 km. This is born in the Eastern Hills as a product of the confluence of the La Osa and Upata streams. It has a total drainage area of 17,536 hectares -12,991 urban and 4,545 in the rural part-, and an average slope of 5.3%.

- ***Tunjuelo River Basin***

According to the reference system of the Agustín Codazzi Geographic Institute (IGAC), the highest elevation of the basin is located at 3,850 m above sea level. n. m. and the lowest, at 2,530 m above sea level. n. m. Along its channel (34.10 km in length within the urban perimeter) three well-defined sectors can be distinguished with respect to its topography and drainage.

30% of Bogotá's population lives in the Tunjuelo basin. The sanitary system is mainly composed of the Tunjuelo Medio, Comuneros Lorenzo Alcatraz and Limas interceptors, and the Grancolombiano, Cartagenita and Isla Pontón San José pumping stations that discharge into the river. In the future, they will deliver the Tunjuelo interceptor
Canoes with final discharge to the Canoas wastewater treatment plant (planned).

- ***Torca River Basin***

The Torca canal basin has a drainage area of 6,008.69 hectares and its main axis has a length of 13.06 km, from the Bosque de Pinos residential complex, located on Carrera 6 and Calle 153, to its delivery to the Bogotá River. This rises in the Eastern Hills and flows into the Torca-Guaymaral wetland system, at the height of the North Highway and near the grounds of the Jardines de Paz cemetery. In turn, it drains north of the middle basin of the Bogotá River.

This basin is made up of three main sub-basins: El Cedro, San Cristóbal and Serrezuela, which begin in the Cerros Orientales, and whose channels are lined.

2. DGREENHOUSE GAS EMISSIONS DIAGNOSTICS AND CLIMATE RISK ASSESSMENT

This chapter is divided into three sections. First, the District GHG Emissions Inventory is presented, updated with data from 2017 and the methodology used to generate it. The second part includes the emissions scenarios obtained for the city, which were built using the Pathways tool, taking into account 2017 as the base year. These correspond to the scenario *business as usual* (BAU) and the ambitious mitigation scenario. Finally, the results of the Climate Risk Assessment (ERC) and the probability and impact of the climate threats analyzed in the Capital District are described.

2.1. Bogotá Greenhouse Gas Emissions Inventory

This section summarizes the update of the city's Greenhouse Gas (GHG) Emissions Inventory for 2017, which covers the emission sources contemplated within the Basic Report of the Global Protocol for Community-Scale Greenhouse Gas Emission Inventories (GPC). **The results showed a total of 11,421,724 tons of CO_2 equivalent (t CO_2It is**).

General information about this report is presented in Table 1 and the subsequent sections detail the results and expand information on the calculation methodology.

Table 1 Basic data of the Bogotá Greenhouse Gas Emissions Inventory

Methodology general employee	The guidelines described in the "Global Protocol for Community-Scale Greenhouse Gas Emission Inventories (GPC)" and the "IPCC Guidelines for the Preparation of National GHG Inventories – 2006" were followed. The CIRIS tool was used for the calculation and reporting. The respective CIRIS file is attached with the inventory of Bogotá for 2017.
Report type	Emissions generated by all sources found within the basic report of the GPC protocol were included. This is the basic report suggested for those cities, such as Bogotá, where agricultural activities do not have a significant contribution to the economy and there are no industries that produce process emissions such as clinker kilns, glass production from carbonates, iron and steel,

	ferroalloys, among others, within the limits.
Process of review of the inventory and results	The inventory was reviewed and approved by C40. Email with approval is attached.

2.1.1. Results

Bogotá's GHG emissions inventory includes the stationary energy, transportation, and waste sectors. For 2017, the total emissions recorded were 11,421,724.32 t CO_2e, with the largest contribution (48%) from the transportation sector. Table 2 shows the results by sector and scope.

Scope 1 emissions in stationary energy and transportation refer to those derived from the consumption of fossil fuels and biomass in these sectors, and scope 2 emissions correspond to those generated by the use of electrical energy. Scope 1, for the waste sector, includes all emissions caused by the treatment or final disposal of solid and liquid waste in Bogotá, in sites located within the city limit, and scope 3 refers to the of city waste that is treated outside.

Table 2 GHG emissions by sector

Sector	Total GHG emissions (t CO2e)

	Range 1	Range 2	Range 3	Total (report basic)
ENERGY STATIONARY	2.892.512	1.019.157	NO	3.911.669
TRANSPORT	5.419.303	130	NO	5.419.433
WASTE	2.090.621	THAT	1	2.090.622
TOTAL	10.402.437	1.019.287		11.421.724

*NA: not applicable. NE: not estimated (emissions not required for the basic report)

Chart 7 shows the percentage distribution of emissions by sector and Chart 8 and Chart 9 disaggregate the results by subsector. Details of the emissions figures by subsector are presented in Table 3.

Graph 7 Percentage distribution of GHG emissions by sectors

WASTE

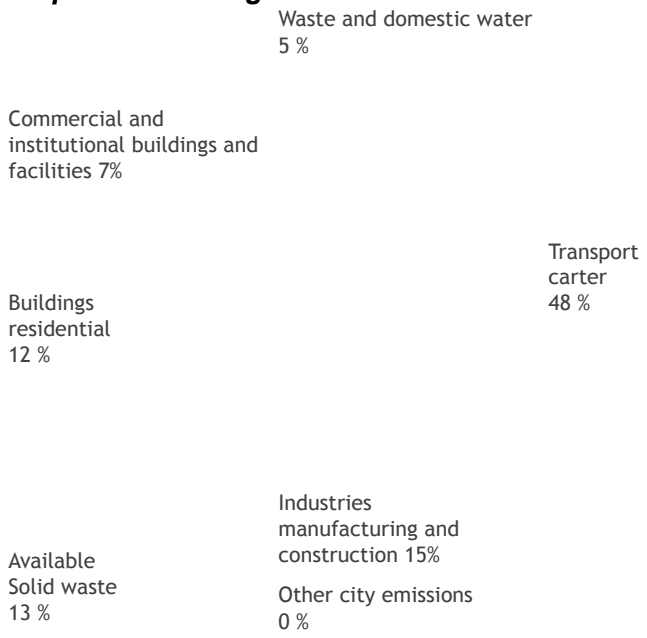
18 %

48 %

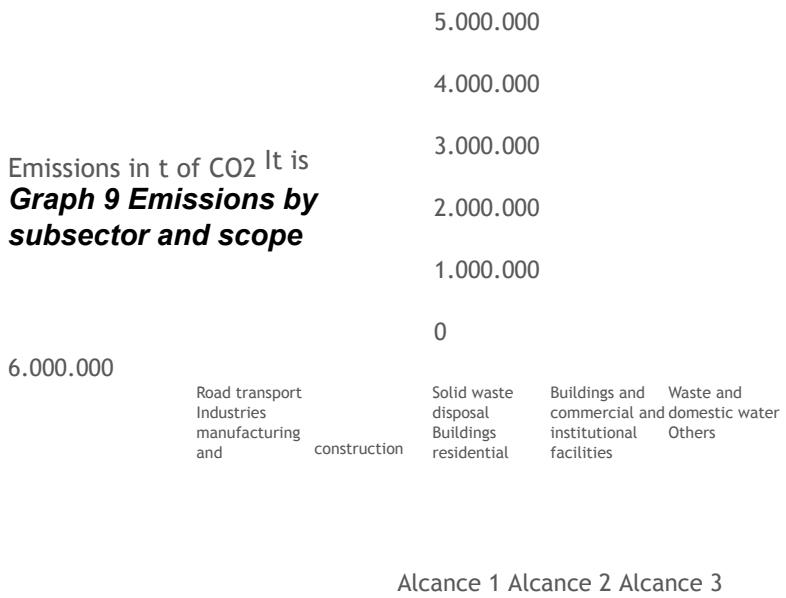
ENERGY STATIONARY 34%

TRANSPORT

Graph 8 Percentage distribution of GHG emissions by subsector



*Other city emissions include: fuel use in the railway and agricultural sectors; incineration and biological treatment of waste, and fugitive emissions from the distribution of natural gas.



Graph 9 Emissions by subsector and scope

*Others include: use of fuels in the railway and agricultural sectors; incineration and biological treatment of waste, and fugitive emissions from the distribution of natural gas.

As seen in Graphs 8 and 9, as well as in Table 3, the main GHG emissions in the city come from road transportation, manufacturing industries, solid waste disposal, residential buildings, and commercial and institutional facilities. These five sectors add 95% of the city's emissions, as explained below:

- Use of fuels (gasoline, diesel, compressed natural gas, biodiesel and ethanol) and electrical energy for road transportation (approximately 47.7% emissions from fuel use and less than 1% from electrical energy consumption).
- Consumption of fossil fuels, biomass and electrical energy in the industrial sector (13% for fuel use and 2% for electrical energy consumption). ● Disposal of solid waste in the Doña Juana landfill, with a contribution of 13% of the city's total emissions.
- Use of fuels (natural gas and LPG) and energy in the residential sector (8.6% and 3.5%, respectively, of the city's emissions).
- Consumption of electrical energy and fuels (natural gas, LPG, charcoal and wood) in the commercial and institutional sectors (3.7% and 3.3%, respectively, of the city's emissions).

The remaining emissions (5%) are generated by the disposal of city wastewater that is not connected to a treatment plant; use of fuels in the railway and agricultural sectors; incineration and biological treatment of waste, and fugitive emissions from natural gas distribution.

Table 3 GHG emissions by sector and subsector (basic report)

No. of	GHG emission source (by sector and subsector)	Total GHG emissions (tonnes CO_2It is)

reference a del GPC		Range 1	Alcance 2	Alcance 3	Total
I	STATIONARY ENERGY				
I.1	Residential buildings	985.472	415.103	NO	1.400.575
I.2	Commercial buildings and facilities institutional	418.464	372.437	NO	790.901
I.3	Manufacturing and construction industries	1.470.567	231.587	NO	1.702.154
I.4.1/2/3	energy industries	NO	NO	NO	-
I.4.4	Generation of energy supplied to the grid	NO	THAT	THAT	-
I.5	Agriculture, forestry and fishing activities	IE	31	NO	31
I.6	Unspecified sources	IE	NO	NO	-
I.7	Fugitive emissions from coal	NO	THAT	THAT	-
I.8	Fugitive emissions from gas distribution natural	18.009	THAT	THAT	18.009
SUBTOTAL	STATIONARY ENERGY	2.892.512	1.019.157	-	3.911.669
II	TRANSPORT				
II.1	Road transport	5.418.632	130	NO	5.418.762
II.2	Railway	671	NO	NO	671
II.3	Maritime navigation	NO	NO	NO	-
II.4	Aviation	NO	NO	NO	-
II.5	off road	IE	IE	NO	-
SUBTOTAL	TRANSPORT	5.419.303	130	0	5.419.433
III	WASTE				
III.1.	Solid waste disposal	1.498.520	THAT	NO	1.498.520
III.2	Biological waste treatment	129	THAT	NO	129
III.3	waste incineration	NO	THAT	1	1
III.4	Waste and domestic water	591.972	THAT	NO	591.972

SUBTOTAL	WASTE	2.090.621	0	1	2.090.622
TOTAL	TOTAL	10.402.437	1.019.287	1	11.421.724

*NA: not applicable. NE: not estimated (emissions not required for the basic report). NO: they do not occur in the city. IE: included in another subsector.

2.1.2. Methodology

Table 4 lists, for each sector, a summary of the information sources of the data used in the calculation and the specific method for estimating GHG emissions.

Table 4. Data used to calculate greenhouse gas emissions

Source of emission of GEI (por sector and subsector)	Activity data and data source	Source of factors issue	Calculation method
STATIONARY ENERGY			
Buildings residential Buildings and facilities commercial and institutional Industries manufacturer as there is construction	Fuels (gas natural, LPG, diesel, gasoline, kerosene and fuel) and energy electricity consumed in the District in the 2017, expressed in TJ, broken down to residential sectors, commercial and industrial. Supplied by the Planning Unit Mining-Energy (UPME)	• Emission factors de CO_2 of the order national associated with the burning of fuels, supplied by the UPME in the FECOC tool (emission factors of fuels Colombians) • Emission factors for CH_4 and $N._2O$: volume 2, chapter 2 of the guidelines of the IPCC 2006 for national inventories of GHG (Tables 2.2., p. 2.17; 2.3., pp. 2.18 and 2.19; 2.4., p. 2.20 and 2.21; 2.5., pp. 2.22 and	• Emissions of range 1: IPCC, 2006, equations 2.1. and 2.2, volume 2, chapter 2. • Emissions of range 2: appendix A, standard corporate of accounting and report - WBCSD, WRI, SEMARNAT

Agriculture, forestry and activities fishing		2.23)	

		The emission factors of CO_2 employees for consume of electricity (range 2) are specific for the system Interconnected National and are published annually for the UPME	
TRANSPORT			
Transport carter	Fuels (gas natural, diesel, biodiesel and ethanol) sold to ground transportation in the District in the 2017, expressed in TJ, supplied by the unit of Mining Planning Energy (UPME)	● Emission factors de CO_2 of the order national associated with the burning of fuels, supplied by the UPME in the FECOC tool (emission factors of fuels Colombians) ● Emission factors for CH_4 and N_2O: volume 2, chapter 3	● Emissions of alcance 1: IPCC 2006, Equation 3.2.1, volume 2, chapter 3 ● Emissions of range 2: appendix A, standard corporate of accounting and report -

Railway	Coal consumption and ACPM by trains in Bogotá, supplied by Turistren	of the guidelines of the IPCC 2006 for national inventories of GHG (tables 3.2.1, 3.2.2 and 3.4.1)	WBCSD, WRI, SEMARNAT
WASTE			
Available waste solids	Amount of waste deposited in the filling, in tons, and characterization of waste supplied by the Subdirectorate of Final Disposition of the Administrative unit of Public Services	k = generation rate of methane, OX = oxidation factor, MCF = factor correction for methane, DOCF= fraction of the DOC that can decompose, F =	IPCC 2006, volume 5, chapter 3, method FOOT them decomposition of first order

	(Uaesp)	fraction of methane in landfill gas. Default data of the IPCC, in the guidelines for elaboration of 2006 inventories, volume 5, chapter 3, tables 3.2, 3.3, 3.5	
Treatment biological waste	Amount of waste treaties and fertilizer organic supplied by the Institute for Social economy (IPES), Selective Route of Organic Waste in Market Squares District	● Factors of emission: data by IPCC default available in the volume 5, chapter 4, table 4.1, Page 4.7 of the IPCC guidelines of 2006 for the inventories national GHG	IPCC 2006, equations 4.1 and 4.2, volume 5, chapter 4

Incineration of waste	Amount of waste hospitable incinerated, supplied by the Subdirectorate of Environmental Control Public sector (Ecocapital report - Infectious Waste) of the Secretariat Environment District	Dry matter content in incinerated waste and burned, fraction of carbon in matter dry, carbon fraction fossil in total carbon, oxidation factor, factor emission of CO2, CH4 and N2O: IPCC, 2006, tables 5.2., p. 5.20; 5.3., p. 5.22; 5.6., p. 5.24	IPCC – 2006, equations 5.1. and 5.3, volume 5, chapter 5
Waters residuals and domestic	Population served through different routes of treatment of wastewater from the city. Fact supplied by the Aqueduct Company and sewage Bogota	● DBO per capita: estimated data from data of entrance to the WWTP Saltpeter ● Emission factor for each system of treatment, Maximum capacity of production of CH4 and factor correction for methane (fraction)	GPC, equation 8.9

		by type of treatment: guides 2006 IPCC, table 6.3	

2.2. Greenhouse gas trajectories

This section presents the emission trajectories based on two scenarios that depend on the implementation of mitigation measures: *business as usual (BAU)*, which shows how emissions would increase if greenhouse gas mitigation actions are not carried out, and the ambitious one that details the level of emissions achieved if the city complies with the commitments made by this Administration: reducing greenhouse gas emissions by 15%. greenhouse gases by 2024, by 50% by 2030 and be carbon neutral in 2050.

2.2.1. Business as usual emissions trajectory

To construct the BAU emissions scenario, the Pathways tool was used, calibrated with the emissions inventory described previously, using assumptions presented below.

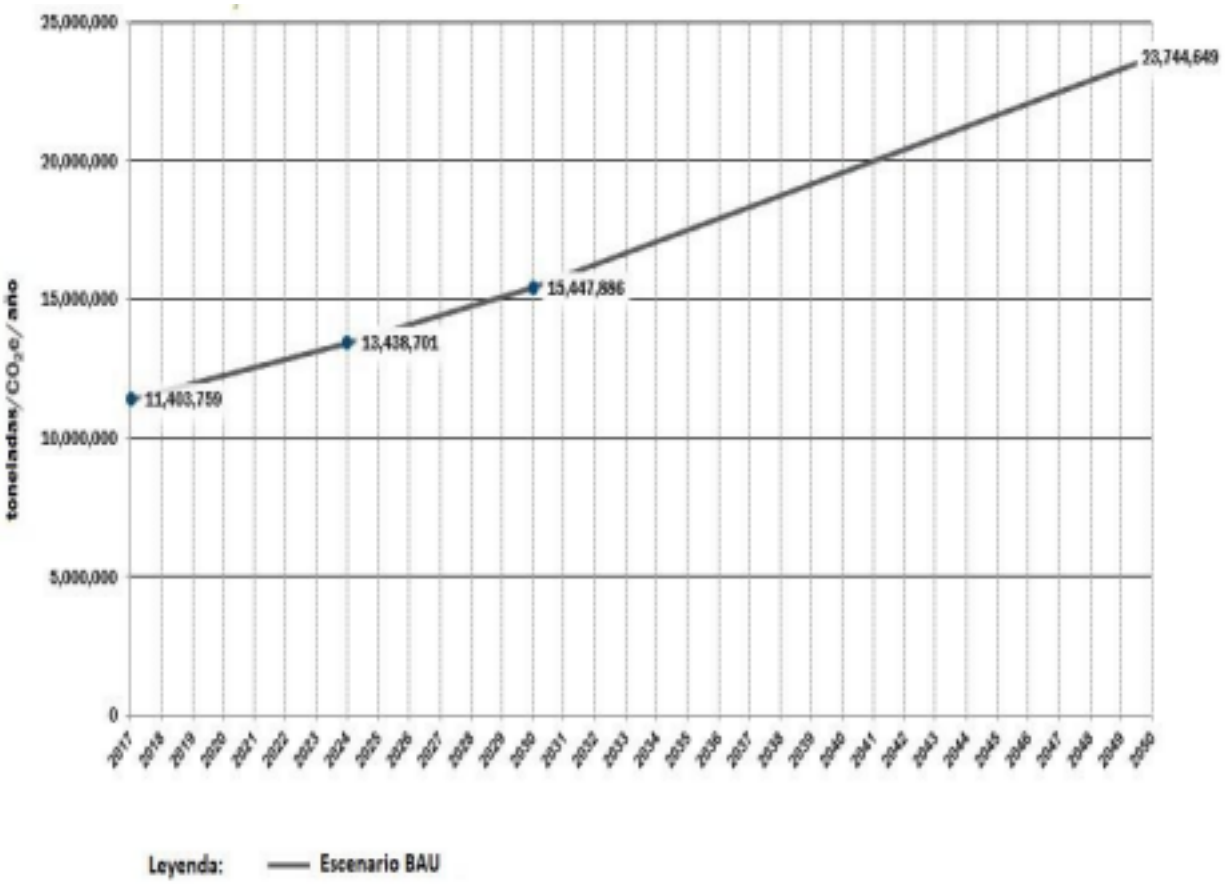
The base year of the projections is 2017. In this year, the city had 7,333,415 inhabitants, according to the information provided by the District Planning Secretariat. The same source estimated a population of 8,034,649, 8,434,700 and 9,164,445 inhabitants in 2024, 2030 and 2050, respectively. With these values, the compound annual growth rate of the population of 1.31%, 0.81% and 0.42% was calculated for the periods 2017-2024, 2024-2030 and 2030-2050 respectively.

The city's GDP in 2017 was $147,080 billion in constant 2005 pesos, according to data reported by the National Administrative Department of Statistics (DANE). The growth rate of this measure was estimated based on information from the city's 2020-2030 medium-term fiscal framework. Thus, a GDP growth of 3.29% was assumed for the period 2017-2024 and 3.75% for 2024-2030.

The Medium-Term Fiscal Framework is updated annually, and the one used for the projection does not include the effects of COVID-19. Once the new fiscal framework is in place, we will seek to update the growth scenario to incorporate the effects of this pandemic. For the period 2030-2050, a constant growth rate was assumed and equal to that reported in the previous document for 2030: 3.7%. Based on this, the city's GDP is expected to increase 25.4% by 2024, 56.4% by 2030 and 223.5% by 2050.

Graph 10 presents the emissions trajectory in the scenario *business as usual* For the city. Emissions in this grow at an average annual rate of 2.37%, 2.35% and 2.17% in the periods 2017-2024, 2024-2030 and 2030-2050, respectively, to double in 2050, with respect to the base year level.

Graph 10 Emissions trajectory in the BAU scenario from 2017 to 2050

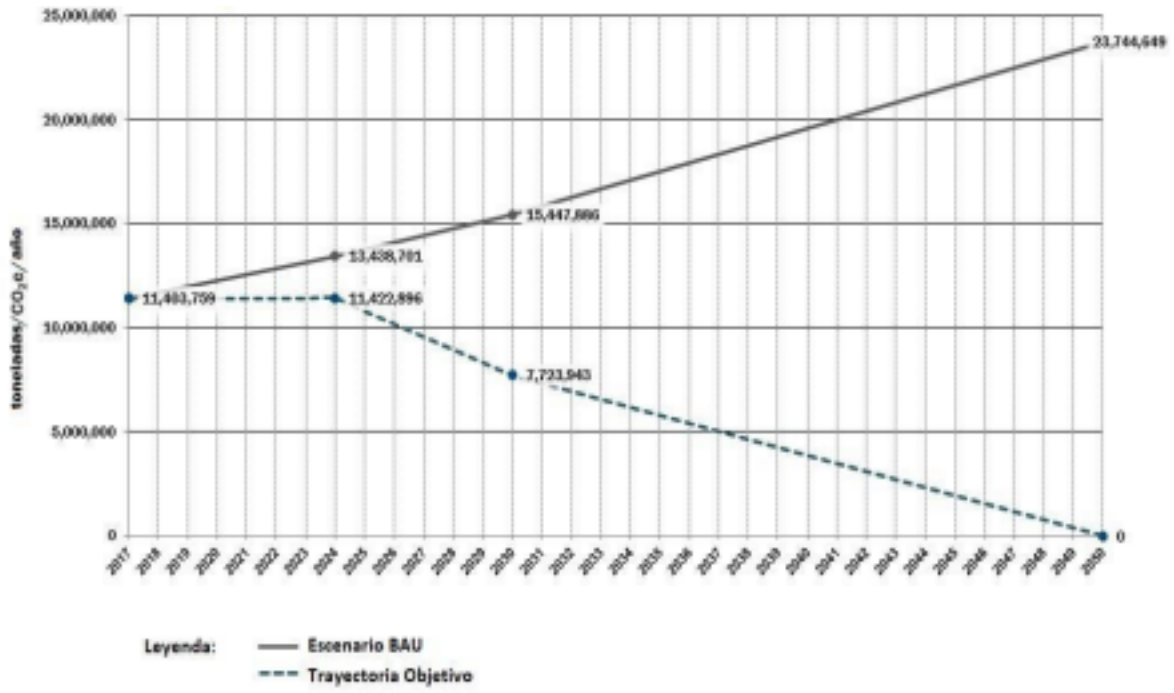


Source: SDA, 2020.

2.2.2. Emissions mitigation trajectory

The city's mitigation objectives are defined in three periods. In this sense, it will work to ensure that, as established in the Development Plan, in 2024 emissions are 15% below the level of the BAU scenario, reaching in said year the same level of emissions as that of the base (2017). In 2030, emissions, with respect to the BAU scenario, are expected to be 50% to obtain a reduction of approximately one third compared to the base year (2017) and, finally, the aim is to achieve carbon neutrality in 2050. Compliance with These goals imply the stabilization of emissions in the period 2017-2024 and the sustained reduction from that year on, as can be seen in Graph 11.

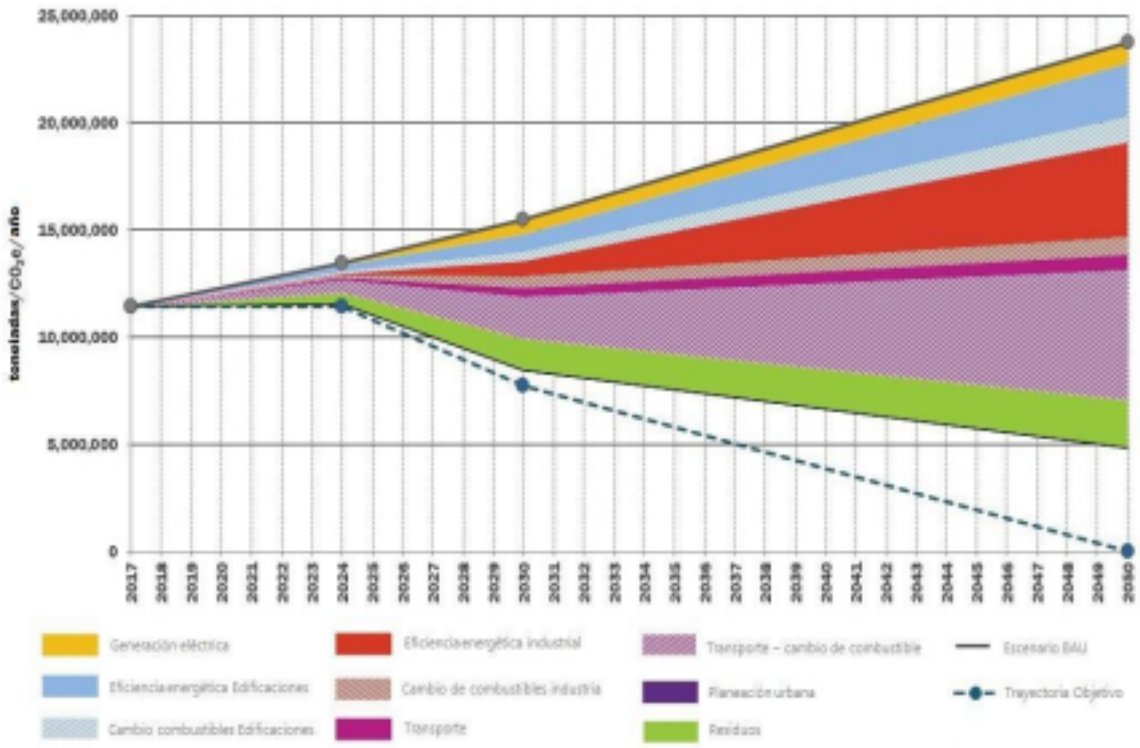
Graph 11 Emissions trajectory in the BAU scenario and emission goals scenario from 2017 to 2050



Source: SDA, 2020.

The ambitious mitigation scenario leads the city to meet the goals set for 2024 and 2030. This would mean that in 2050 a decarbonization of 80% would be achieved, as can be seen in Graph 12. The required mitigations would be achieved with actions implemented through increased energy efficiency, improvements in the disposal and treatment of solid and liquid waste and the electrification of a large part of final energy consumption. Our scenarios assume that the country will maintain the provision of low-carbon electricity, because the responsibility for the decarbonization of this form of energy falls on the national territory. The greatest mitigation potential comes from fuel substitution in all final consumption sectors, especially in transportation.

Graph 12 Emissions trajectory in the ambitious mitigation scenario 2017 to 2050



Source: SDA, 2020.

The decarbonization of electricity generation in the national interconnected system and distributed generation with renewable energies reduces 8.8% and 5% of total emissions in 2030 and 2050, respectively. Energy efficiency and fuel substitution measures in commercial and residential buildings lead to mitigating 17% and 19% of total emissions in 2030 and 2050, in that order. Likewise, these types of measures in the industrial sector manage to mitigate 17.1% and 27.6% of total emissions for 2030 and 2050, correspondingly.

The transport sector is, as indicated above, the one with the highest emissions in the city. Modal shift and fuel substitution measures - mainly electrification - reduce emissions by 37% both in 20°30 and in 2050. The aggregate mitigation potential in the waste sector is 20.2% in 2030 and 11.6% in 2050.

Tables 5, 6, 7, 8 and 9 present the mitigation potentials of the groups of actions, modeled in the ambitious scenario (the characterization of the actions of

mitigation that were considered for the construction of the ambitious emission scenario is presented in chapter 4 of this document).

Table 5 Mitigation potential in electricity generation actions

Generación de electricidad						
	2024		2030		2050	
	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción
Energías renovables distribuidas	2,801	<1%	81,054	1.1%	203,102	1.0%
Descarbonización del suministro eléctrico	110,785	5.7%	635,560	8.4%	809,807	4.2%
Sub-total	113,586	5.8%	716,614	9.5%	1,012,909	5.2%

Table 6 Mitigation potential in energy actions in buildings

Energía en edificaciones						
	2024		2030		2050	
	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción
Nuevas construcciones - eficiencia	163,100	8.3%	607,077	8.0%	2,334,269	12.2%
Modenización - Medidas arquitectónicas	0	0.0%	97,041	1.3%	283,745	1.5%
Enfriamiento de áreas - eficiencia	244	<1%	900	<1%	1,362	<1%
Usos térmicos - eficiencia / cambio de combustibles	120,472	6%	501,150	7%	820,434	4.2%
Iluminación - eficiencia	63,321	3.2%	151,800	2.0%	187,756	<1%
Equipamiento - eficiencia	37,124	1.9%	55,687	<1%	83,530	<1%
Alumbrado de calles - eficiencia	10,592	<1%	8,840	<1%	22,119	<1%
Sub-total	394,854	20.2%	1,422,495	18.8%	3,743,214	19.3%

Table 7 Mitigation potential in industrial actions and other stationary energies

Industria y otras energías estacionarias						
	2024		2030		2050	
	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción
Industria - cambio de combustible	94,569	4.8%	566,839	7.5%	818,624	4.2%
Industria - eficiencia	64052	3.3%	698,850	9.2%	4,459,753	22.9%
Sub-total	158,621	8.1%	1,265,689	16.7%	5,278,378	27.2%

Table 8 Mitigation potential in transportation actions

Transporte	2024		2030		2050	
	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción
Cambio modal - bicicleta / caminar	40,017	2.0%	149,591	2.0%	282,028	1.5%
Cambio modal - transito	40,017	2.0%	231,532	3.1%	393,024	2.0%
Vehiculos de motorizados - eficiencia / cambio de combustible	645,091	33%	2,017,195	27%	6,100,391	31.4%
Vehiculos de carga - eficiencia / cambio de combustible	NI	NI	NI	NI	NI	NI
Otras reducciones en transporte	50,000	2.6%	280,000	3.7%	385,000	2.0%
Sub-total	775,125	39.6%	2,678,228	35.4%	7,160,443	36.8%

Table 9 Mitigation potential in waste actions

Residuos	2024		2030		2050	
	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción	t CO2e/año	% total de reducción
Reducciónde residuos	0	0.0%	0	0.0%	0	0.0%
Reciclaje	62,630	3.2%	223,860	3.0%	457,980	2.4%
Compostaje	139,990	7.1%	433,230	5.7%	934,080	4.8%
Tratamiento de aguas residuales	12,537	<1%	515,069	6.8%	558,088	2.9%
Otras reducciones asociadas a residuos	301,275	15.4%	301,275	4.0%	264,006	1.4%
Sub-total	516,432	26.4%	1,473,434	19.5%	2,214,154	11.4%

23. Climate Risk Assessment (ERC) and probability and impact

According to the Intergovernmental Panel on Climate Change (IPCC), the risks of climate change are concentrated in cities, where there is a high density of people and economic activities (IPCC, 2014), which which makes them fundamental actors for the development of ambitious and transformative climate actions, which allow the objectives of the Paris Agreement to be met and promote the fair transition towards low-carbon, competitive and climate-resilient societies.

Under this context, the Climate Risk Assessment (ERC) represents a useful instrument to identify and evaluate the current and future climate risks associated with the climate threats that arise in the Capital District, by making the scale and severity of their impacts visible; provide evidence on its distribution in the territory and the elements with the greatest exposure - emphasizing vulnerable people and communities - and provide the tools for the development, prioritization and implementation of adaptation and resilience actions (C40, 2018).

To address the impacts of climate change, Bogotá has the District Risk and Climate Change Management System (SDGR-CC), created through

Agreement 546 of 2013, from which disaster risk planning and management instruments have been developed, through the District Disaster Risk and Climate Change Management Plans (PDGRDCC). The most recent Plan has an implementation horizon of 2018-2030 and

was approved by the District Council for Risk Management and Climate Change through Agreement 001 of November 9, 2018 and adopted by Decree 837 of 2018.

Until now, this instrument is responsible for the planning of the SDGR-CC, which organizes its activity and generates synergy with the other processes of environmental, territorial and development planning in the Capital District. However, and despite the fact that climate risks are incorporated into its programs, strategies and goals, the PDGRDCC has greater emphasis on disaster risk management and its implementation has been developed with an emergency focus; Therefore, the District Administration, through the 2020-2024 District Development Plan "A New Social and Environmental Contract for 21st Century Bogotá", proposes a new strategy for planning effective actions to reduce risks and adapt to the climate crisis, through the Climate Action Plan 2020-2050.

In this regard and considering the circumstances and capabilities of the city, as well as the legal framework that regulates disaster risk management and climate change management at the national and District level, the methodology of the Action Planning Framework was adjusted. C40 Climate Strategy to generate the ERC, as the technical support document to identify adaptation priorities and define actions that reduce the level of climate risk and increase the adaptive capacity and resilience of ecosystems, infrastructure and communities in the District Capital.

Taking into account the above, this section describes, first of all, the regulatory macro of climate change management, focused on adaptation and resilience in the international, national and District context. Then, the methodology used for the development of the ERC is addressed, which describes aspects such as the conceptualization of the Climate Risk Index for Adaptation to Climate Scenarios (CRI). Subsequently, the sensitive recipients/sectors that are strategic for the analysis of the impact of climate threats are described. Next, the regional approach of the ERC is presented, its differential aspects compared to other analyzes in Latin America and the limitations specific to the Capital District, which must be addressed in subsequent analyses.

Regarding the ERC results, the future trends of temperature and precipitation projected to 2040 are shown; a synthesis of the intensity and frequency of climate threats in Bogotá, as well as in the regional context, including their materialization in conditions of climate variability (Boy and Girl Phenomena). Then, the CRI maps and the impact of climate threats on sensitive receptors/sectors are presented and analyzed. Finally, the adaptation priorities of the Capital District and the main strategies proposed to increase its adaptive capacity and resilience are described.

2.3.1. Normative Framework

In the international context, adaptation and resilience are developed mainly through three instruments: the Paris Agreement, the 2030 Agenda and its 17 Sustainable Development Goals (SDGs) and the Global Summit on Climate Action.

The Paris Agreement was approved by Colombia through Law 1844 of 2017 and in that same year the National Climate Change Policy (PNCC) was adopted, which includes in its objective, reducing the risks of climate change and taking advantage of the opportunities. that it generates.

With regard to the 2030 Agenda, the country has recognized that the SDGs are the integrating axis of the different development agendas that it is currently advancing, such as the Paris Agreement and the Final Peace Agreement, by allowing the coherent alignment of both actions public and private "around a common objective" (DNP, 2018, page 9); For this reason, the national government formulated the "Strategy for the Implementation of the Sustainable Development Goals (SDG) in Colombia" (CONPES 3918 of 2018), as the reference framework for the formulation of public policy guidelines and strategies for the implementation of the 2030 Agenda.

Regarding adaptation to climate change, SDG 13: *Climate Action*, establishes the need for countries to strengthen resilience and adaptive capacity to climate-related risks and natural disasters; Likewise, SDG 11: *Sustainable Cities and Communities*, points out the urgency for cities and human settlements to adopt and implement policies and plans to, among other issues, promote adaptation and resilience to the risks generated by climate change, in line with the Sendai Framework for Risk Reduction of Disasters 2015-2030.

The Global Summit on Climate Action is the scenario established by the United Nations (UN) to boost ambition and accelerate actions for the implementation of the Paris Agreement. The Summit was held in 2019, in which 77 countries - including Colombia - more than 100 cities and nearly 100 business leaders announced important commitments to combat the climate crisis (UN, 2019).

At the national level, the regulatory framework for climate risk management, adaptation and resilience is developed through Law 1931 of 2018, "by which guidelines for the management of climate change are established", for the purposes of which, Among others, it is to reduce the vulnerability of the population and ecosystems to the effects of climate change.

Regarding disaster risk management, there is Law 1523 of 2012, "By which the national disaster risk management policy is adopted and the National Disaster Risk Management System is established and dictates other provisions", which establishes an approach to addressing risks that evolved beyond disaster management, to a management and resilience approach, recognizing climate change as a reality that the country faces, which requires development of measures to reduce vulnerability and improve resilience, in response to observed or expected changes in the environment

climate and its variability. This Law was accepted by Bogotá, through Agreement 546 of 2013.

Law 1931 of 2018, in its Article 9, establishes that municipal and district authorities must incorporate climate change management into their development plans and territorial planning plans, as well as in other planning instruments, taking the Plans as a reference. Comprehensive Sectoral Climate Change Management Plans, which, in the case of the Capital District, corresponds to the Comprehensive Climate Change Management Plan Bogotá-Cundinamarca Capital Region (PRICC). In response to the above, the ERC of Bogotá integrates a regional approach that considers the materialization of the climate threats analyzed in the Administrative and Special Planning Region (RAPE), made up of the departments of Cundinamarca, Meta, Tolima and Bogotá and the Metropolitan Region, which includes Bogotá and 33 municipalities, 30 in the jurisdiction of Cundinamarca, 2 in Huila and 2 in Meta.

Furthermore, Article 11 of the aforementioned Law establishes the need for articulation and complementarity between the processes of adaptation to climate change and disaster risk management, which must lay its foundations in the aspects related "to the processes of knowledge and "reduction of the risk associated with hydrometeorological and hydroclimatic phenomena and potential modifications in the behavior of these phenomena attributable to climate change." The above includes both the Comprehensive Territorial Climate Change Management Plans, the Departmental and Municipal Risk Management Plans, as well as the other planning instruments defined in Chapter III of Law 1523 of 2012.

At the District level, Bogotá has been making progress in incorporating climate change into territorial development. In 2015, the reform of the Emergency Prevention and Attention Fund -FOPAE- was carried out, and the District Institute for Risk Management and Climate Change -Idiger- was created, which undertakes actions and generates guidelines for risk management. of disasters and adaptation to climate change, within the framework of the coordination of the SDGR-CC.

At the level of planning and management instruments, as mentioned, the Capital District has the District Disaster Risk and Climate Change Management Plan for Bogotá, 2018-2030, which "guides actions for knowledge of disaster risk , of climate change and its effects, risk reduction, mitigation and adaptation to climate change, and emergency and disaster management. The Plan is structured into components, objectives and programs - which include strategic lines of action - and goals. These components are: 1. Knowledge of disaster risk and climate change and their effects: 2. Climate change mitigation; 3. Disaster risk reduction and adaptation to climate change: 4. Emergency and disaster management and 5. Governance for response to emergencies and disasters.

As can be seen, the current Plan has a greater emphasis on disaster risk management and although it integrates strategic lines and goals that seek to reduce climate risks and increase adaptive capacity, the city requires an instrument that clearly defines the priorities in adaptation issues, articulate them with disaster risk management, climate change management and

environmental and integrates and incorporates them as a fundamental issue in the territorial ordering and planning processes.

2.3.2. Territorial and regional approach of the ERC

The ERC was carried out based on a differential territorial approach, which considered specific indicators for the urban and rural areas, taking into account the particular territorial dynamics (environmental, social, geographical, economic and cultural) that are presented in them, therefore , the units of analysis for the urban area were defined as the Zonal Planning Units (UPZ) and for the rural areas of Bogotá the Rural Planning Units (UPR). The rural/urban differentiation facilitates the prioritization of actions at the local level that respond to the particular characteristics of the territories, as well as their needs, vulnerabilities and implementation facilities.

On the other hand, the city's ERC integrates as a fundamental aspect the regional approach as the articulating axis of the planning and territorial ordering processes in the context of the climate crisis. For this purpose, it incorporates the historical and current analysis of climate threats in the context of the Administrative and Special Planning Region (RAPE) and the Bogotá-Cundinamarca Metropolitan Region, which makes their frequency and impact visible, as well as decision making. articulated to reduce risk and implement transformative actions to adapt to climate change.

However, the absence of regional indicators for most threats only allows us to analyze their projection against the climate trends included in the Third National Communication on Climate Change (TCNCC), without involving exposure or vulnerability analysis. Despite the above, the regional approach is considered strategic, to define climate actions that increase resilience and adaptation capacity from a perspective of comprehensive and synergistic territorial planning and ordering.

2.3.3. Methodological approach

The methodological approach used for the ERC of Bogotá integrates a quantitative and qualitative approach, which adapted the methodology of the C40 Climate Planning Program to the regulatory and territorial context of the city, as well as its capacities and resources. It is made up of twelve (12) steps that had to be addressed online and in parallel, to obtain the expected results. This sequence is shown in graph 13.

Graph 13. Stages of the CKD of Bogotá D.C. 3.

Selection of

1. set the probability

and the

Conceptualization of risk

climate

7. Historical analysis and current of the

indicators for estimating the

CRI

8. Analysis of future trends 2040

6. Preparation of IRC maps

9. Analysis of IRC maps

11. Evaluate the different impacts related to

4. Management

cartographic

5. Array construction and

2. Identification of threats

10. Determine sensitive sectors and

magnitude of impact

12. Evaluate the adaptive capacity

2.3.3.1. Climate Threats analyzed

The historical analysis of the occurrence of disasters triggered by extreme climate events in Bogotá, as well as by possible changes in the future climate such as increases or reductions in precipitation and temperature, allowed us to identify the following climate threats that occur most frequently in Bogotá. the city and that were included in the ERC:

- Floods.
- Mass movements.
- Forest fires.
- torrential avenues.
- Urban heat islands.

In addition to the above, the ERC integrates an analysis of historical trends of the minimum temperatures that have occurred in the city, based on the information available in the databases of the SDA and Idiger weather stations, since the year 2001 to 2020.

2.3.3.2. Conceptualization of the Risk Index for Adaptation to Climate Change Scenarios (IRC)

The estimation of climate risk in the Capital District was carried out based on the Risk Index for Adaptation to Climate Change Scenarios (IRC), understood as a quantitative approach that integrates a series of social, economic, environmental, infrastructure indicators, among others. , which allow evaluating the components of climate risk (climate threat factors, exposure and vulnerability - sensitivity

and adaptive capacity-) in a differentiated manner in urban and rural contexts, considering the projected change in temperature and precipitation variables to 2040.

As shown in Table 10, their values are classified as high, medium and low for the CRI of floods, mass movements and urban heat islands. In the case of torrential floods, only presence-absence is recorded, while the threat is classified in such a way that in Bogotá the CRI for forest fires was classified as very low, low, medium, high and very high.

Table 10 CRI evaluation scale

IRC	Avenues Torrential	Floods	Mass movement s	fires forestry	heat islands urban

Very high	N.A	N.A	N.A		N.A
High	N.A				
Half	N.A				
Low	N.A				
Very low	N.A	N.A	N.A		
Bit probable	N.A	X	X	N.A	N.A
Presence	X	N.A	N.A	N.A	N.A
Absence	X	N.A	N.A	N.A	N.A

Source: IDIGER, 2020

According to the methodology of the C40 Climate Action Planning Program, the general formula of the CRI results from the interaction between hazard factors, exposure and vulnerability (sensitivity/adaptive capacity), as shown below .

However, the conditions of Bogotá, being a city with hillside and savanna geoforms, where the distribution of periphery-center vulnerability is irregular and heterogeneous, led to an adjustment to the CRI formula proposed by C40, with the so that its results represent the way in which the threatening phenomena of the city physically materialize and that, in addition, it will integrate the climatic incidents that have occurred in the last 18 years.

This modification generated greater weight to the threat component of the IRC formula and allowed its value to be adjusted, according to the way in which climate threats have historically manifested themselves in the Capital District at the level of UPZ for the urban area and UPR for the rural areas of Bogotá. .

For this purpose, a variable called SIRE Factor (Information System for Risk Management and Climate Change) was incorporated into the IRC formula for climate threats: floods, torrential floods, mass movements and urban heat islands. The final IRC formula used in Bogotá is:

$$\text{\textbf{\textbf{\textbf

In addition to the IRC, the following indices were calculated, which are an essential part of determining the impact of climate threats in the Capital District:

- Exposure Index for Adaptation to Climate Change Scenarios, which was crossed with the coverage of the location of the vital combined sanitary sewer networks, storm sewer network, electrical network, road network and natural gas, to determine its possible impact on the climate threats analyzed.

- Sensitivity Index for Adaptation to Climate Change Scenarios, which indicates the degree to which the population is affected by threats

analyzed, by integrating indicators that show conditions of vulnerability, such as socioeconomic, demographic aspects, housing conditions, among others.

- Adaptive Capacity Index, whose indicators focused on the calculation of distances from homes to essential equipment such as hospitals, fire stations, etc., internet connection and income, among other aspects that contribute to reducing the impacts of threats. climatic.

- Total Vulnerability Index for Adaptation to Climate Change Scenarios, which is obtained from the proportionality between the sensitivity and adaptive capacity factors and represents an estimate of the physical, economic, social, environmental or institutional susceptibility or fragility of suffering. adverse effects in the event that a physical event associated with a hydroclimatological phenomenon occurs.

The definitions of the IRC components were aligned with the national regulatory framework for disaster risk and climate change management as follows:

The threat, in the terms of Law 1523 of 2012, is understood as the latent danger that a physical event of natural origin, or accidentally caused or induced by human action, occurs with sufficient severity to cause loss. of lives, injuries or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service delivery and environmental resources.

The threat values were obtained from the plans of the proposed update of the Territorial Planning Plan -POT- prepared within the framework of National Decree 1077 of 2015, multiplied by the percentage of change in the climatic triggers, that is, the projection of the temperature and precipitation variables to 2040 (maximums and minimums). The indicators included for the threat component correspond to the area of the selected analysis units (UPZ, and UPR), which is at risk (high, medium and low) for floods, urban heat islands, forest fires, torrential floods and mass movements.

For its part, exposure refers to the presence of people, means of subsistence, environmental services and economic and social resources, cultural assets and infrastructure that, due to their location, may be affected by the manifestation of a threat (Law 1523 of 2012). . This component allows us to know the degree to which a system is exposed to events associated with climate change. For this reason, its indicators relate the extension of vital elements such as the road network, equipment, storm sewage, telecommunications, etc. , as well as the number of homes and people in medium and high threat.

Vulnerability is obtained from the proportionality between the sensitivity and adaptive capacity factors and indicates the physical, economic, social, environmental or institutional susceptibility or fragility of suffering adverse effects in the event that a physical event associated with a hydroclimatological phenomenon occurs. .

The sensitivity component allows the identification of population groups, ecosystems, infrastructure, essential public service equipment, etc. with greater vulnerability to climate threats, as well as socioeconomic variables that make the population more fragile to the effects of climate change; while, the adaptive capacity, interpreted for the purposes of the ERC as resilience, in accordance with Law 1931 of 2018, refers to the capacity of social, economic and environmental systems to face a dangerous event, trend or disturbance, responding or reorganizing themselves so that they maintain their essential function, their identity and their structure, while preserving the capacity for adaptation, learning and transformation. In general terms, this component considers a set of resources and local conditions that support or limit the ability of a system to adapt or respond successfully to changes in climate.

The indicators used in each component are listed in Annex 1. Likewise, in the ERC technical support document, the methodology used for the normalization of the indicators is specifically addressed, which allowed them to be made comparable to each other (i.e. that is, taking them to a scale from 0 to 1), to subsequently carry out the

operations that will yield the results of threat, exposure, sensitivity and adaptive capacity of the CRI for each climate threat analyzed.

2.3.3.3. Methodology to determine probability and impact

The analysis of the probability and impact of the climate threats that occur in Bogotá was carried out in two stages. As a first measure, based on the qualitative approach proposed in the C40 Rapid Climate Risk Analysis Methodology, work sessions were held with experts in knowledge, prevention, care and response to emergencies and disasters from Idiger, the SDA, the CAR of Cundinamarca and the EAAB, in which, under a participatory approach, the probability of climate threats was evaluated, understood as the frequency with which these are expected to occur by 2040, as well as the impact, defined as the anticipated consequences for people, assets, or services when the event occurs, whichever experiences the greatest impacts (such as flooded homes, disrupted transit service, etc.).

Both the probability and impact assessment were carried out considering an IPCC RCP 8.5 climate scenario, that is, in the assumption that no actions are implemented to reduce GHG emissions and these continue to increase throughout the 21st century.

The evaluation carried out by the experts was contrasted and adjusted based on the results of the IRC, which made it possible to include the geographical context and the historical analysis of the materialization of climate threats in the Capital District and thus obtain a more precise evaluation of their probability. and impact. The values of these variables were included in the C40 Climate Actions Selection and Prioritization (ASAP) process, as a criterion to determine the primary benefits generated by adaptation actions to reduce climate risk and increasing adaptive capacity and resilience to the impacts of climate change.

2.3.3.4. Method for projecting climate scenarios

The climate scenarios (temperature and precipitation projections to 2040) for Bogotá were obtained from the study generated by the Institute of Hydrology and Environmental Studies (Ideam) for the Third National Communication on Climate Change in 2015, in which the “Joint Reliability Average Ensemble Method” (REA) that combines the outputs of general circulation models, taking into account two criteria: performance and convergence; and also, from the multi-scenario ensemble that averaged the changes evidenced by the 4 RCPs (2.6 – 4.5 – 6.0 – 8.5) for each period (2011-2040, 2041-2070 and 2071-2100). For this reason, the result shown by the scenarios, both in temperature and precipitation, corresponds to an average of the RCPs and not to one in particular.

2.3.4. Receivers/sensitive sectors analyzed

The sensitive recipients or sectors correspond to the elements (people, livelihoods, ecosystems, essential services - sewage networks, energy,

telecommunications, etc. -; infrastructure; or economic, social or cultural assets) that, due to their location, are or can be affected by the manifestation of climate threats[2].

Its selection was made based on the strategic lines for disaster risk management at the level of public and private entities defined in Law 1523 of 2012 and in Decree 2157 of 2017, by which general guidelines are adopted for the preparation of the disaster risk management plan for public and private entities. Based on these elements, the impact of the climate threats included in the ERC is analyzed. In the case of Bogotá, the following sensitive recipients/sectors were selected:

- Public services-lifelines.
- Buildings and constructions.
- Transport.
- Ecosystems and biodiversity.

2.3.5. Differential aspects of the ERC of Bogotá.

The ERC of Bogotá differs from other similar documents analyzed for Latin America, since:

- It proposes an articulation between territorial planning instruments such as the Territorial Planning Plan (POT) and risk management such as the District Plan for Disaster Risk Management and Climate Change 2018-2030.

[2]The definition of sensitive receptors/sectors was adapted from the C40 Climate Action Planning Framework methodology, based on the exposure definition of Law 1523 of 2012.

- It develops a regional commitment as a fundamental aspect in the planning and territorial ordering processes in the context of the climate crisis. ● The analysis was carried out on a UPZ and UPR scale, which allows adaptation actions to be focused, considering the rural and urban differences and needs of the city.

2.3.6. Limitations of CKD

Regarding the limitations of the ERC, the first thing is to point out that the analysis presented does not correspond to a risk analysis in the definitions of the national regulatory framework for risk management (Law 1523 of 2012), since the spatial and temporal scale is not They adjust to the requirements required in these cases, nor do they represent or substitute a technical concept of risk, therefore, in the ERC of Bogotá the IRC is adopted, as noted above.

Regarding the methodological aspect, the main limitation was related to the adjustment of the methodology of the C40 Climate Planning Framework to the context and circumstances of Bogotá, since the general risk formula proposed by C40 does not

integrate the realities physical characteristics of the city (hillside and savanna areas that configure a particular scenario for the manifestation of hydroclimatic events), which caused some variables to have greater representativeness and yield erroneous results that do not correspond to the way in which they have historically materialized. climate threats in the Capital District. This situation involved carrying out a different approach to the formula, including additional variables as mentioned in section 2.3.2, based on a heuristic analysis.

In addition to the above, the selection of indicators for the components of the CRI formula was conditioned by the availability of information at the UPZ and UPR level, the latter being the ones with the greatest information deficit in aspects relevant to the calculation of the sensitivity; such as the proportion of people with storm and sanitary sewer coverage, access to essential services, telecommunications, among others. This condition forced us to assign values of zero (0) to these areas, which results in a lower sensitivity or proportion of exposed elements (e.g. service networks), which may not represent the territorial reality.

Specifically, the analysis of the climate threat from forest fires only covers the rural area of the District, since the threat map that the city currently has only includes 4.2% of the urban land. Despite the above, the analysis of the historical and current situation was carried out based on the data record for the entire Capital District.

Regarding the projection of information of climate scenarios, a treatment of urban centers and a regional subscaling is required, which corresponds to the territorial realities, since the projection of national scales can have a distorting effect in UPR such as the Eastern Hills, where the confluence of two precipitation regimes (monomodal and bimodal) may not be adequately represented.

The information available to analyze extreme winds, food or water shortages does not have a repository of regional or District information that allows addressing these dimensions in the ERC, however, a first approach to the analysis of the probability and impact of water scarcity was carried out. in an IPCC RCP8.5 scenario based on work sessions with EAAB experts, which made it possible to identify climate actions that contribute to reducing the effects of said threat. Climate risk analyzes on the risk dimensions defined in the TCNCC: food security, water resources, biodiversity and ecosystem services, health, human habitat and infrastructure, were included as goals of the PAC's adaptation and resilience actions.

In line with the above, the availability of information to integrate biological risk phenomena into the CKD at the UPZ and UPR scale requires implementing institutional actions, especially with the District Health System, therefore, associated indicators were not considered in this analysis. with said variable. As in the previous case, the analysis in the health risk dimension of the TCNCC was included as a goal of the adaptation and resilience actions of the PAC.

Finally, for the most part, the indicators used in this analysis do not have

information available at a scale suitable for territorial planning purposes such as 1:5000, therefore, the IRC maps are considered indicative maps that support urban planning. but they do not constitute environmental determinants under the terms of Law 388 of 1997.

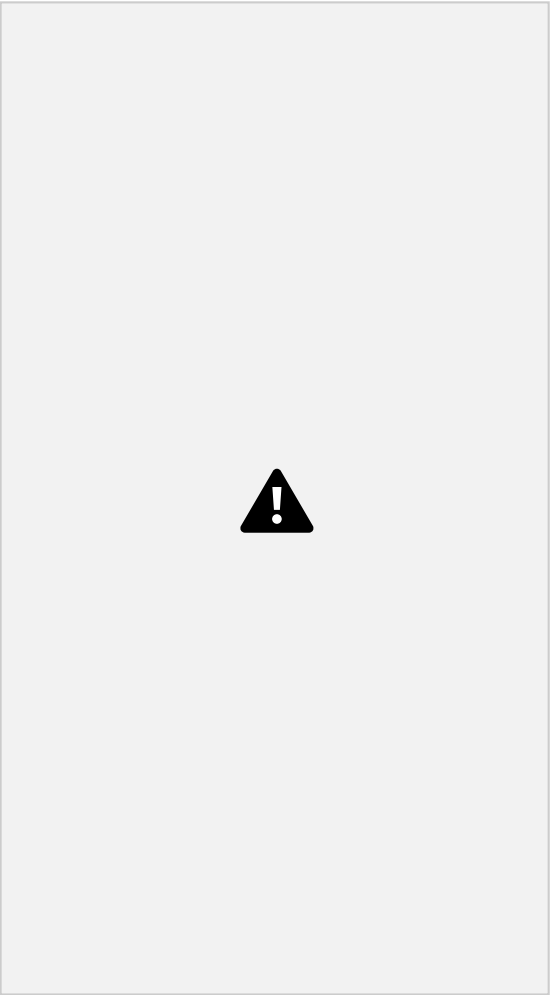
2.3.7. Results and analysis

2.3.7.1. Climate Scenarios

According to the projection of temperature and precipitation scenarios in the period 2011-2040 of the Ideam, with reference to the average temperature recorded between 1986 and 2005 and the annual rainfall record between 1975 and 2005, the characteristics of urban and rural areas of Bogotá will influence an increase or decrease in these two variables in a differentiated manner, as shown in graph 14.

Regarding the variation in average annual temperature, it is expected that by 2040 it will increase by 0.25 °C in rural areas and 0.65 °C in urban areas. With respect to precipitation, increases are also expected mainly in the percentage of change, but, as with temperature, its values are not standard for the entire Capital District, due to the different orographic characteristics of the territory. In this way, while to the west of Bogotá it is predicted that the percentage of rain will increase by up to 35% by 2040, in the Eastern Hills and the southern area of the town of Sumapaz a significant decrease of up to 15% is expected.

Graph 14 Scenario of increase in average annual temperature in degrees Celsius (Right) and scenario of change in rainfall in percentage (Left)



Fountain: Ideam, 2017.

2.3.7.2. Climate threat analysis

The analysis of climate threats covers the following components:

1. Intensity and frequency: in which the historical and current record of the occurrence of climate threats is analyzed. In the regional context, depending on the availability of information, information is included for the Administrative and Special Planning Region (RAPE) and the Metropolitan Region. The maps that accompany the analysis were obtained from information from the TCNCC and the PIRCC, Goevisores of the Government of Cundinamarca, institutional portals of the Ideam, the UNGRD, and information from the CARs of Cundinamarca, Corporinoquia, Cormacanera, and CorpoGuavio. These maps are listed in Annex 2.

At the District level, the information is analyzed differentially for the rural (UPR) and urban (UPZ) areas, emphasizing the intensity and frequency of the threat analyzed, according to the data reported on the SIRE platform and information from the SDA. This section includes the analysis under conditions of climate variability (Niña-Niña phenomena).

2. Analysis of future climate trends: carried out based on the projection of Ideam precipitation and temperature scenarios in 2017.

3. IRC analysis: which describes the priority UPZ and UPR in terms of adaptation to climate change.

4. Analysis of the impact on sensitive receptors/sectors: in which emphasis is placed on the elements (public services, buildings, road network, ecosystems and communities) with greater exposure to climate threats, as well as their respective effects.

Below, the points indicated for the climate threats included in the ERC are developed, organized in order of priority for adaptation to climate change in the Capital District, that is: floods, mass movements, forest fires, torrential floods and islands of urban heat.

2.3.7.2.1 Floods

According to the Idiger[3], floods are caused by excess water, invading areas that are dry under normal conditions. This phenomenon plays an important role in the regulation of water systems, for this reason, when these systems are modified or areas susceptible to flooding are occupied, they can cause damage.

Floods may occur due to overflow or flooding (on roads, hailstorms, increase in river flow, etc.), depending on the conditions in which they occur. For the IRC analysis, both the flood threat map due to overflow and flooding were used. However, it is specified that the factors that increase the threat of flooding due to overflow are associated with strong precipitation events due to climate variability and the unregulated management of rivers for different uses of water (agriculture and livestock, drinking water supply , electric power generation, basin transfer processes[4], etc.); while, flooding, involves the deficits of the city's rainwater and/or mixed network to drain runoff water.

[3]At https://www.idiger.gov.co/rinundacion

[4]Regarding the transfer of basins, in Bogotá D.C., the import of water from the Orinoquia could generate surpluses over the conditions of the natural landscape, overflowing its capacity for hydraulic regulation. Therefore, to face the risks of climate change, it is necessary to establish hydraulic management conditions that consider climate change scenarios.

2.3.7.2.1.1 Intensity and frequency analysis

2.3.7.2.1.1.1 Regional context

In the territories that make up the RAPE, the historical occurrence of floods occurs with greater intensity in the months of March and May, particularly in the Andean region and in some departments near Cundinamarca such as Tolima. In departments like Meta, the materialization of floods, given its hydrography -
belonging to the Orinoquia-, are associated with erosion processes and droughts, mainly in the months of June-August and March-May, therefore, for the most part, this department presents low probabilities of threat from flooding with the exception of the municipality of Puerto Lopez. The above occurs in a similar way in the department of Boyacá, where low levels of risk from flooding occur in most of its municipalities (RAPE, 2016).

In the metropolitan context[5], south of the Province of Sabana Centro[6]are the areas with the highest risk of flooding. The bodies of water of the Bogotá River, the Muña Reservoir and the Tominé Reservoir converge in these territories, which increases the threat to this type of events. The TCNCC projections indicate that the municipalities of Cota and Chía are those with the greatest climatic risk due to flooding. To which is added Tocancipá, which, according to the analysis of the historical flood events presented in 2010, 2011, 2012 and 2020, is considered a priority intervention area in terms of floods.

On the other hand, in the Province of Sabana Oeste[7], two significant flooding events have occurred: in the village of Santa Cruz, in the municipality of Madrid in 2011 and in the San Antonio rural school in the municipality of Bojacá in July of the same year (SAGA, 2020). In the municipalities of Mosquera and Rosal, flooding events have occurred that must be considered, in order to adequately manage natural buffers such as the Gualí Wetland.

The Province of Soacha presents a medium risk of flooding, associated with the presence of the Bogotá River. However, in this area, between 1981 and 2011, the greatest number of floods occurred, after Bogotá. For its part, the Guavio Province, according to CAR reports, is the area of the Metropolitan Region that has suffered the most. for these types of events. Data from the Government of Cundinamarca reports that, with the exception of La Calera and Guatavita, all municipalities in the area have been affected by floods.

[5]In this document, the proposal for the Metropolitan Region of Bogotá is made, made up of 45 municipalities, which incorporate those included in the PRICC, as well as supply areas of the Bogotá Aqueduct and Sewer Company-EAAB, electricity generation areas managed by said company. company and integration projects for rail transport.

[6]It includes the municipalities of Cajicá, Chía, Tabio, Cogua, Tenjo, Cota, Tocancipá, Gachancipá, Zipaquirá, Nemocón and Sopó.

[7]It includes the municipalities of Madrid, Mosquera, Subachoque, Zipacón, El Rosal, Facatativá, Funza and Bojacá.

Regarding the Provinces of Oriente and Sumapaz, there is not enough information to analyze historical trends due to flooding; However, some records indicate that in the areas of Fómeque and Fusagasugá there have been floods that have caused damage to its inhabitants.

Likewise, the lack of information on floods in the Meta Sumapaz and Meta-Chingaza areas does not allow for an analysis of historical trends of this type of events. However, it is important to note that these territories will present one of the highest water deficit indices in the region, according to the TCNCC. In the Meta-Chingaza area, projections show that there will be a reduction of up to 15% in precipitation in years of El Niño phenomenon (Ideam, et al., 2017).

On the other hand, the frequency of floods reported in the PRICC, for the period 1980 to 2011, indicates that the municipalities of the Bogotá River and Magdalena River basin are the ones that have been most affected by this type of events; The municipalities of the savannah, Bogotá, Soacha, Chía, Cajicá, Tocancipá and Zipaquirá and the lower area of the Bogotá River (Girardot, Tocaima, Apulo, Ricaurte and Viotá) were the ones with the highest occurrence. In Magdalena Medio; Puerto Salgar, Guaduas and Caparrapí are the areas in which flood events occur the most and in the plain foothills, Medina is the municipality that has suffered the highest frequency of these events (Ideam et al., 2012a).

2.3.7.2.1.1.2 District Context (rural and urban)

In the case of Bogotá, flooding due to waterlogging and in some sectors due to overflow, is the most frequent climatic threat historically. In the rural area of the Capital District, in the strip of the Eastern Hills in the south of Usme, it usually rains between 1,000 and 1,200 mm/year, being rainier in a good part of the town of Santafé and in the east of Chapinero and San Cristóbal. . In this sector, when the trade winds from the southeast intersect with ascending air masses from the Savannah to the east, and are accompanied by sufficient humidity provided by the Orinoquía, the collision of the two air masses gives rise to the formation of cumulonimbus-type clouds with great vertical development, which suggests a high probability of significant torrential rains. In these scenarios, there is a potential increase in flooding in the towns of San Cristóbal, Usaquén, Chapinero.

On the other hand, in the northern area of the town of Sumapaz, which is the rainiest in the Capital District, strongly influenced by the humidity from the Orinoquía, the average annual rainfall reaches up to 1,500 mm, which, in scenarios of climate change, allows estimating increases in precipitation that could lead to floods in the middle and lower part of the Tunjuelo basin, as already occurred in 2002.

In the urban area, the Capital District has a probability of flooding due to the overflow of the main tributaries of the Bogotá River: the Fucha, Salitre and Tunjuelo rivers due to problems derived from the floods in these basins. Regarding flooding, as seen in graph 15, the year 2010 was the year in which this type of events occurred the most with nearly 1,400, followed by 2012, in which approximately 800 were recorded. In 2019, in comparison with the period included

Between 2013 and 2018, there was a slight increase, reaching around 250 flooding events. However, the potential for flooding has a patchy distribution in the city, in which cases such as Galerías or Cedritos occur, areas in which inadequate management of the sewage system occurs.

Graph 15 Flood events (2010-2020)

Source: (SIRE, 2020)

In addition, it is estimated that 6,719 people and 30% of the urban area are in medium, high and low flood threat; The towns of Bosa, Kennedy, Engativá, Fontibón, Suba, Tunjuelito, Rafael Uribe Uribe, Usme, Ciudad Bolívar and San Cristóbal being the most affected and those that coincide with the towns that present the highest rates of poverty, population density and that are located near wetlands, river banks and mountain slopes.

The above is related to aspects such as deforestation, waterproofing of the soil with materials such as concrete and cement, inadequate discharge of liquid and solid waste that clogs drains, filling of wetlands (natural flood zones) and constructions in rounds of rivers, among other factors, that make flooding the climatic threat to which Bogotá must adapt with greater urgency.

The analysis of the Niño-Niña climate variability conditions (graph 16) indicates that between 2001 and 2019, in the years with Niña events (moderate and strong), the highest number of flood emergencies occurred in the Capital District, especially in the months of April and November. As an example, in 2011, 1,849 flood emergencies were recorded; That year a strong La Niña event occurred..

Number of events presented

fear of flooding[n]

Graph 16 Flood events (2010-2020)

weak child
500

ANALYSIS OF PAST TRENDS
FLOOD THREAT

2000

strong boy
1500

Moderate girl
0
2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016
2017 2018 Year of occurrence

Source: Idiger, 2020

strong girl

Moderate Child
1000

This behavior is corroborated with the analysis of precipitation data obtained by three Ideam meteorological stations in the Capital District (Botanic Garden, Venado de Oro and El Dorado Airport), in a period of 39 years, which indicates that the maximum accumulated precipitation It was presented in 2011, coinciding with a La Niña Fuerte phenomenon.

2.3.7.2.1.2 Future analysis

At the regional level, the La Niña phenomenon is expected to be even stronger with increases in rainfall of more than 40% in the next 20 years, which allows for more frequent and intense flooding, especially in the provinces of Guavio, Sabana Centro and Western Savannah. In the latter, the municipalities of Nemocón, Cogua, Chía and Cota stand out as those with priority intervention in terms of adaptation to climate change due to flooding events. In sectors such as Soacha and in the north of the Sabana,

particularly in the municipality of Tocancipá, stronger and more intense rains are also expected.

In the Capital District, the climate scenarios for 2040 indicate that the localities in the west of the city will have the most difficult conditions for managing the climate threat due to flooding. The precipitation projections in said period coincide with the area of the town of Kennedy, in which the greatest historical events of flooding and flooding have occurred, particularly in the UPZ Calandaima, in which, during the last decade, 94 have occurred. flooding events and 31 due to flooding, precisely where the El Burro Wetland is located, and other wetland areas. The above indicates the need to continue and strengthen the development of actions aimed at recovering the water connectivity of the wetland and increasing local permeability rates in this area of the city.

Likewise, the entire town of Bosa, which despite having a relatively dry regime, in relation to other points in the city, turns out to be one of the most sensitive and exposed sectors to the effects of climate change; considering that many of the homes located there are below the flood levels of the Tunjuelo River (Islandia, Sauces, Independencia, Echeverri, San Diego neighborhoods), and the Tibanica wetland (Charles de Gaulle neighborhoods, La Esperanza de Tibanica), which constitutes a high risk of flooding. In this area, an increase

of 40% of the volume of precipitation by 2040, could have a cumulative effect when the Tunjuelo River floods coincide, between the months of May and July (due to the influence of the Orinoquia on the Sumapaz), generating an extraordinary event. of flooding under climate change scenarios.

In the rural area, the town of Sumapaz presents total multi-year precipitation averages that indicate a condition of relatively low values towards the southwest of said town, oscillating between 900 and 1,000 mm. These values would increase between 11% and 30% in 2040, which can trigger flood events in said area. Cases such as that of the UPR Norte and the Thomas van der Hammen Reserve, which according to the historical analysis of the records of the Ideam, SDA and Aeropuerto meteorological stations, has the highest rainfall in Bogotá and which additionally records one of the lowest temperatures of Cundinamarca, is projected to present between a 15% to 40% increase in precipitation, which implies a greater requirement in water management in the zonal plans of the sector.

2.3.7.2.1.3 CRI analysis

Exposure to service networks turns out to be one of the strategic aspects in the analysis of CRI due to flooding. In this sense, the Flood Exposure Index for Adaptation to Climate Change Scenarios (graph 17) shows that the Eastern Hills and the Tunjuelo Basin (lower and upper part) are the areas in which the networks, for example, storm sewers, are mostly exposed to flooding in climate change scenarios. To the north of the city, the axis of Carrera 7 is configured as a priority intervention area, considering its high vulnerability.

Graph 17 Flood exposure index for adaptation to climate change scenarios Bogotá D.C.

Source: Idiger, 2020

In this way, the areas of the city that present the highest CRI values due to flooding, as seen in graph 18, are located to the west and correspond to the towns of Suba, Engativá, Bosa, Kennedy and Fontibón.

Graph 18 CRI due to flooding

Source: Idiger, 2020

The results at the UPZ and UPR level are shown in table 11. As can be seen, in the urban area, the UPZ Tibabuyes (Suba), Engativá (Engativá), Patio Bonito (Kennedy), Galerías (Teusaquillo), Bosa Occidental (Bosa), Calandaima (Kennedy), El Porvenir (Bosa), El Rincón (Suba) and Granjas de Techo (Fontibón), present risk

climate due to overflow and flooding; For its part, the UPR Northern Zone in the town of Suba is configured as a priority intervention in matters of flooding under climatic scenarios in the rural area of the Capital District.

With respect to overflow events, the UPZ Tunjuelo (Tunjuelo), Bosa Central (Bosa) and Zona Franca and Granjas de Techo in Fontibón represent the areas of the city in which adaptation actions should be prioritized.

Table 11 UPZ and UPR with higher CRI

UPZ	LOCATION	OVERFLOW/ASSOCIATED BODY OF WATER	PONDING
TIBABUYES	BELOW	Bogotá River Special Management Area Channels and tributaries Tibabuyes and Conejera Wetland	X
TUNJUELITO	TUNJUELITO	Tunjuelo River Chiguaza Gorge Hoya del Ramo ravine Quebrada Santa Librada Yomasa Gorge Canal San Vicente I	
BOSA CENTRAL	BOSA	Tunjuelo River Humedal Tibanica	
ENGAGEMENT	ENGAGEMENT	Bogotá River Special Management Area Marantá Canal Jaboque Wetland	X
BEAUTIFUL PATIO	KENNEDY	Bogotá River Special Management Area Muiscas Canal (38th Street) Cundinamarca Canal Interceptor	X
GALLERIES	TEUSAQUILLO	Canal Negro Salitre river	X

BOSA OCCIDENTAL	BOSA	Tunjuelo River	X
CALANDAIMA	KENNEDY	Bogotá River Special Management Area Fucha River	X

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		El Burro Occ Wetland – El Burrito Canal Alsacia Canal Magdalena Canal Castilla Cundinamarca Channel	
FUTURE	BOSA	Bogotá River Special Management Area Cundinamarca Channel	X
FREE ZONE	FONTIBON	San Antonio Canal or central canal Fucha or San Cristóbal River	
THE CORNER	BELOW	Salitre River Tibabuyes Wetland Wetland Arm- Corinto Suba Transversal Avenue Canal Canal Cafam	X
ROOF FARMS	FONTIBON	Fucha or San Cristóbal River San Francisco Canal Hayuelos Canal Canal San Antonio or Canal Central	
RURAL PRIORITY			

UPR NORTH ZONE	BELOW	Bogotá River Special Management Area (Suba-Cota and Guaymaral) Guaymaral Wetland Canals and Fences	X

Fountain: Idiger, 2020

2.3.7.2.1.4 Impact Analysis

- Public services and lifelines: UPZ 87 (Tintal), 86 (El Porvenir), 82 (Patio Bonito), 79 (Calandaima), 78 (Tintal Norte) and 77 (Zona Franca), as well as the UPZ

52 (La Flora), 55 (Diana Turbay), 54 (Morocco), 60 (Parque Entrenubes), 62 (Tunjuelo) and 63 (El Mochuelo), located at the transition of the Cerros Orientales and the Tunjuelo system, correspond to the areas with greater exposure to floods and in which there will be greater impacts on the three types of networks analyzed as part of the recipient/sensitive sector public services and lifelines: storm sewage, sanitary and electrical network (graphs 19, 20 and 21) .

Storm sewer networks are the most exposed to flooding (graph 20) and therefore, on which the greatest impacts can occur. The Sewerage Management Units - UGAS that present the greatest threat in flood scenarios are those related to the Tunjuelo basin in the lower and upper part. In the north of the city, the axis of Carrera 7 is strategic for the development of actions to reduce the impact of flooding.

Graph 19 Exposure of the storm sewer network to floods *Source: Idiger, 2020*

Graph 20 Exposure of the combined sanitary sewer network to floods *Fountain: Idiger, 2020*

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Regarding the electrical network (graph 21), the effects can occur in the UPR Río Sumapaz, Rio Blanco and Río Tunjuelo. For its part, the UPR Northern Zone and Cerros Orientales present a medium and low threat due to impacts on its electrical network.

Graph 21 Exposure of the electrical network to floods

Source: Idiger, 2020

- ***Buildings:*** Bogotá does not have high levels of rainfall, if compared to other cities in the country (average annual rainfall of 800 mm), however, every year, in the winter season the city floods and this is because urban developments have hardened the soil altering its natural hydrological cycle.

- ***Transport:*** When flooding events occur in Bogotá, one of the sectors most affected is transportation. This is due to the flooding processes on the roads, which, in many cases, are a product of inadequate maintenance of the rainwater drainage channel networks, or due to high levels of precipitation that exceed their drainage capacity.

The towns of Bosa, Kennedy and Suba are those with a high threat due to damage to the road network, while the towns of Usaquén, Chapinero, San Cristóbal, Ciudad Bolívar, Usme, Santa Fe and Candelaria have a medium threat. to effects on the road network. In the rural area, the Río Tunjuelo, Cerros Orientales and Río Blanco UPRs have the greatest impact on the road network in the event of flooding events.

- ***Ecosystems and biodiversity***: Ecosystems play a fundamental role in climate resilience to flood events. However, when heavy rainfall occurs, it can affect trees, different species of fauna, physicochemical quality of the water resource, among others, impacts.

  The town of Sumapaz, which has the largest rural area in the Capital District and houses the largest moor in the world, has a low CRI due to flooding, which is associated with the few anthropogenic interventions and the ecosystem richness that this town presents. However, the behavior of the IRC in the UPR Zona Norte, Cerros Orientales and Río Tunjuelo differs compared to the town of Sumapaz. In these areas, there are high anthropogenic interventions (illegal settlements, pressure on land use), which generate alterations in the dynamics and habitat of ecosystems and, therefore, effects due to flooding events.

2.3.7.2.2 Mass Movements

In Bogotá, the combination of areas with slopes greater than 20° located in areas with constant rain, the lack of drainage networks, the lack of protection on cut slopes, pre-existing quarries, interventions on slopes, water discharges and The absence of stabilization in invasion zones has been the trigger to produce mass movements.

2.3.7.2.2.1 Intensity and frequency analysis

2.3.7.2.2.1.1 Regional context

At the regional level, the RAPE departments are conducive to the development of mass movements due to characteristics associated with relief, precipitation, and anthropogenic actions, which facilitate the materialization of this type of threat. In the last 20 years, the department of Cundinamarca has presented the largest number of mass movements with a total of 678; In contrast, in Meta there have been 54, placing it as the RAPE department with the least occurrence of this type of events.

For its part, in the Metropolitan Region, mass movements have occurred dispersed throughout the provinces and municipalities and are associated with aspects such as the concentration of urban centers, the presence of mountainous elevations and heavy rains. Thus, for example, in Sabana Oeste, in 2018, a landslide occurred affecting 4 tertiary roads, 1 local aqueduct, 7 homes and several crops. This same situation occurs in the

Province of Sabana Centro, in which the landslide that occurred in 2014 stands out as a relevant event, which affected communication between the municipalities of Tabio and Subachoque, causing heavy vehicular traffic and damage to two homes. .

In the Province of Soacha, on the border with the Capital District, the presence of abandoned quarries in the area of Altos de Cazucá and El Divino Niño, have materialized landslide events classified as falling clothes and collapse of surfaces. Since 2006, these types of situations have reported fatalities. For its part, the Guavio Province is classified by the Colombian Geological Service as having a high and very high threat due to mass movements (Colombian Geological Service, 2017); During 2015, in the municipalities of Ubalá and Gachalá, rockfalls occurred that resulted in damages to homes, crops, roads and the death of one person.

Finally, in the Province of Sumapaz, in 2018, the materialization of a mass movement event caused damage to the roads of the tertiary network of the municipality of Pandi, the Caracol road. This resulted in lack of communications between the Santa Helena, Buenos Aires Alto and Bajo villages, and the Guacanonzo village. Likewise, the departmental road between Pandi and San Bernardo was affected at the height of the La Batea sector, where there was a total loss of the road in a 20-meter section.

2.3.7.2.2.1.2District Context (urban and rural)

In the rural area, the towns of Chapinero and Santa Fe present a high threat due to mass movements, particularly the sectors surrounding the Quebrada La Vieja. At the UPR level, the UPR Northern Zone presents conditions such as the absence of slope stabilization and protection measures, as well as the lack of mechanisms for the management of surface and subsurface runoff waters, which make it a favorable scenario for present mass movements.

In the UPR Río Blanco and Río Sumapaz, during the winter wave of 2018 (June-August), mass movements occurred that affected important road sections, generating an emergency in the town of Sumapaz that strongly affected the inhabitants of the two basins due to mobility restrictions (IDIGER, 2019).

In the Tunjuelo River UPR, the exploitation of quarry material causes deep excavations that lead to erosion processes and therefore increase the probability of mass movements occurring. Furthermore, the location of substandard homes, which present deficiencies in their construction and poor practices in wastewater management, generate conditions conducive to the materialization of this type of events, which, added to the social vulnerability of its population, means that the impacts are more serious.

In the UPR Cerros Orientales, they constitute one of the areas in which the greatest mass movements occur, due to aspects such as the presence of illegal settlements in hillside areas, inadequate management of surface waters, intrinsic

characteristics of the terrain (slope, geological formations), adaptation of slopes and quarrying activities; which indicates the need to implement recovery and restoration measures in the area, thus reducing the risk index due to mass movements.

On the other hand, the mass removal in the town of Sumapaz is very subject to the dynamics it presents; One of the most common causes for mass movement to occur are topographic processes, such as areas with very steep slopes where the angle of repose of a material is exceeded. Other causes are the frequent rainfall that occurs in the winter season and anthropic processes such as deforestation, agricultural expansion and poor management of surface and subsurface runoff waters.

Finally, it is worth noting that, in November 2020, as a result of the heavy rains that occurred in the capital, a landslide occurred in the town of Usme that affected the road that leads to the Curibital village, as well as some homes. of the sector. In that same month, at kilometer two via Quiba, an emergency occurred due to mass movements in which 95 families were affected.

In the urban context, the localities most affected by mass removal processes are San Cristóbal, Usme, Suba, Rafael Uribe and Ciudad Bolívar, corresponding to 38.8% of the urban area. It is estimated that 3,550,693 people are located in areas threatened by mass removal. The above is due to conditions such as the construction of settlements in sites previously dedicated to the exploitation of quarries, in fills or slopes that were not technically built and even due to their location in the area of the ravines.

Among the mass movements in the Capital District, the one presented on April 14, 2017 stands out, which unleashed an emergency event inside the National Park, between the towns of Chapinero and Santa Fe. At the site, a landslide of earth material that is located in the northern part of the Park and whose possible triggering factor was associated with the water saturation of the slope; added to the heavy rains that occurred in the indicated period.

The analysis of the conditions of Niño-Niña climate variability indicates that, during Niña events, mass movements originate with greater intensity. During 2010 and 2011, when a strong La Niña phenomenon occurred, land saturation occurred, which led to the instability of slopes, soil and rocks, causing damage to homes, people, equipment, infrastructure and service networks. Public.

However, the occurrence of mass movement processes may be weakly correlated with the years in which climate variability phenomena occur, since the origin of many of these events is anthropic, added to the instability of the terrain. Furthermore, the materialization of this type of events is also

due to household flows from illegal water connections with hoses, which are highly susceptible to leaks that saturate the land and can destabilize the slopes. Likewise, wastewater is dumped directly onto the land, because there is no sewage system. ; eroding natural surfaces and modifying the morphophysiology of the soil.

Hence the importance of configuring, within the monitoring of events due to mass movements, processes to verify drainage conditions in areas with slope instability, since inadequate management of household drainage, added to frequent and extreme rains in the area of The confluence of the Eastern Hills and the Tunjuelo Basin can increase the probability of mass movement processes occurring.

2.3.7.2.1.2 Future analysis

As mentioned, mass movements are influenced by precipitation events that, under certain conditions, generate soil instability; Therefore, the analysis of future trends is carried out based on the change in precipitation levels for the year 2040 in Bogotá.

The projections of change in precipitation for 2040 indicate that in the western part of the city, there will be increases in precipitation levels in relation to its historical average, which added to the projected trends in the change in uses and occupation of land, which respond to needs such as the improvement of road connectivity, service coverage and the migration of inhabitants towards areas of high non-mitigable risk (Carrizosa Umaña, 2012), will increase the occurrence of mass movements in said area.

2.3.7.2.1.3 CRI Analysis

The areas of the Capital District with the greatest exposure to mass movements are located on the margins of the Eastern Hills and in the towns of Usme, San Cristóbal and Ciudad Bolívar, as seen in graph 22. The above, among other aspects, is determined by the orographic conditions of the soil, as well as the economic and social activities developed in said areas.

Graph 22 Index of exposure to movements in nothing for adaptation to climate change scenarios

Fountain: *Idiger, 2020*

When analyzing the adaptive capacity index and the sensitivity index, the areas classified as having a high value of exposure to mass movements have a low sensitivity and a high adaptive capacity, as occurs in the town of Chapinero and the Sumapaz PNN, which It means that in that area the effects of climate change due to the materialization of mass movements are smaller. The above is related to the environmental characteristics of these areas, which provide numerous ecosystem services that mitigate the effects of mass movement events.

In the case of the UPZ located on the margins of the Eastern Hills, which present an index of adaptive capacity that is not favorable to climate change, as a result of anthropogenic interventions carried out in the landscape ecosystem, the effects of mass movements are greater.

Graph 23 shows the results of the CRI for mass movements. As can be seen, the towns of Usme, Ciudad Bolívar and San Cristóbal have the largest number of UPZs with high CRI. These results are summarized in table 12, which indicates the UPZs that are considered priority intervention in terms of adaptation to climate change in the face of the materialization of mass movements.

Graph 23 CRI due to mass movements

Fountain: *Idiger, 2020*

Table 12 UPZ with Higher CRI due to Mass Movements

UPZ	Location
The Flora	use me
San Cristobal North	Usaquen
Brown Blonde	Chapinero
Lucero	Bolivar City
July 20th	San Cristobal
Quiet	San Cristobal
Ismael Perdomo	Bolivar City
Entrenubes Park	use me
San Isidro - Patios	Chapinero
The treasure	Bolivar City
Commoners	use me
Diana Turbay	Rafael Uribe Uribe
The liberators	San Cristobal
Danube	use me
Jerusalem	Bolivar City
Saint Blaise	San Cristobal

Source: Idiger, 2020

2.3.7.2.1.4 Impact analysis

In the rainy seasons, mass movements occur as a consequence of the saturation of the land, which generates instability in slopes, soil and rocks; These conditions lead to damage to homes, people, equipment, infrastructure and public service networks, as described below.

- ***Public services and lifelines***: The UPZs that have the greatest exposure to impacts on storm sewage due to mass movements, which is considered essential for the

collection of rainwater in areas with low permeability, are UPZ 40: Ciudad Montes, UPZ 16: Santa Bárbara, UPZ 38: Restrepo, UPZ 12: Toberín and UPZ 23: Casa Blanca Suba, with the towns of Usaquén and Suba being the ones where the greatest impacts may occur.

Regarding the impacts due to mass movements on the electric energy service, these can be presented in UPZ 38: Restrepo, UPZ 81: Gran Britalia and UPZ 29: Minuto de Dios, belonging to the towns of Antonio Nariño, Kennedy and Engativa.

On the other hand, as part of the analysis of the impact of this type of events on public services, it is worth noting that the Doña Juana Sanitary Landfill, which provides services for the final disposal of waste from Bogotá and other municipalities; has been the subject of landslide events, generating health emergencies in

the city. The last one occurred in April 2020, as a result of the inadequate treatment of leachate and the compaction of waste. For this reason, the impacts of mass movement events on the collection and final disposal of solid waste are high, if extreme rainfall events that can destabilize the landfill terraces are considered; Therefore, it constitutes one of the strategic processes in climate risk management in the Capital District.

- ***Buildings***: Among the main assets affected by the materialization of mass movements are crops and homes, which, in some cases, require the relocation of families immediately, in order to avoid tragic situations that lead to damage to the integrity of people.

  The supersaturation of the soil due to poor management of runoff waters puts infrastructure near seasonal or permanent bodies of water at risk and, in many cases, is the reason for the erosive action of surface and subsurface runoff waters on the slopes of natural cuts, which triggers and/or reactivates local mass removal processes.

  Furthermore, the construction of irregularly shaped housing in the Cerros Orientales, Ciudad Bolívar, Usme, San Cristóbal and in the town of Sumapaz, where social groups with high levels of social vulnerability are settled, such as displaced or migrant populations, means that In these areas of the city there are greater impacts on buildings due to the materialization of mass movements, with serious implications on the well-being and quality of life of the population.

- ***Transport***: This sensitive receptor is one of the most affected by the occurrence of mass movement events, causing damage to the roads and interrupting movement, leaving some areas of the city incommunicado. The areas that present a high threat in the road network are located in the towns of Usme and San Cristóbal, as well as in UPZ 69 and 70 of the town of Ciudad Bolívar. It is worth noting that the TransMiCable of Bogotá, located in the town of Ciudad Bolívar, has a high impact due to mass movements.

- ***Ecosystems and biodiversity***: As mentioned, the areas with the highest CRI due to mass movements are located towards the Eastern Hills and to the south of the city, in the towns of Sumapaz, Usme and Ciudad Bolívar, where there are strategic ecosystems for the Capital District and the region, such as Sumapaz, which is home to 43% of the largest paramo complex in the world. When mass movements occur in these places, the habitat, biological corridors and the stability of ecosystems are affected.

2.3.7.2.3 Forest fires

2.3.7.2.3.1 Intensity and frequency analysis

2.3.7.2.3.1.1Contexto regional

At the regional level, in the period 1980-2010, in the municipalities surrounding Bogotá such as La Calera, Soacha and Choachí, the greatest number of forest fires occurred, as well as in municipalities with slopes and high pressure due to land use. such as Facatativá, Subachoque, Tabio, Cajicá, Cogua and Nemocón. Comparatively with other types of events (floods, landslides and windstorms), forest fires present the largest number of records in the regional context (Ideam et al., 2012a).

The occurrence of forest fires in municipalities adjacent to Bogotá increases the probability that they will also occur in the Capital District, because as with other types of events, they do not meet political-administrative limits; The fire follows the route of the fuel, in this case, the vegetation and, easily, a fire that occurs in a neighboring municipality can easily reach the Bogotá territory. An example of the above was the forest fire reported in the Sumapaz Natural Park in early February 2020, presumably within the town of Sumapaz in the Capital District. However, at the time of providing the response, it was evident that it was an event that originated in the municipalities of Guamal and San Luis de Cubarral (in the department of Meta) on the border with Bogotá. The effects of the fire affected the population of the town of La Unión (town of Sumapaz), thanks to the fact that the smoke (and its particulate matter) quickly reached this sector. Although the fire did not affect homes, it did cause damage to the paramo ecosystem that seeks to be protected under the figure of a National Natural Park.

On the other hand, fire events that occur in the Chingaza National Natural Park (which covers seven municipalities in Cundinamarca and four in Meta) put the water supply for the city at risk, since this protected area is home to the main source of water supply for Bogotá.

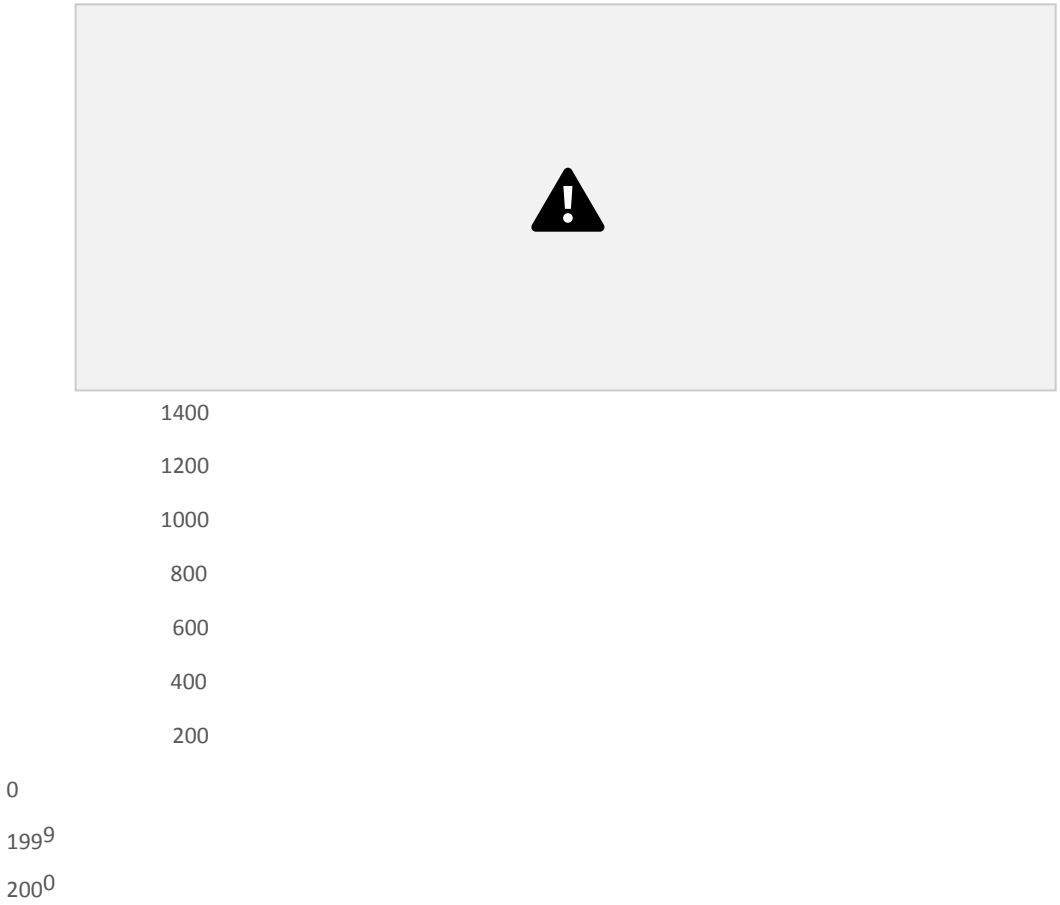
2.3.7.2.3.1.2 District context (rural and urban)

As reported by the District Commission for the Prevention and Mitigation of Forest Fires (CDPMIF), the occurrence of forest fires in Bogotá is associated with the bimodal regime of the city, that is, they occur especially in times of low rainfall and greater temperatures; this is mid: December, March, July and September. However, the first time of the year is usually the strongest or most marked and February is the month in which, generally, the most forest fires occur. In any case, there is no direct relationship between the number of events and the affected area.

An analysis of forestry events (burns, attempts and forest fires) that occurred in Bogotá in the period from 1999 to September 2020 (graph 24), shows that the largest number of burnings was recorded in 2001; In the case of attempts, it was 2009 and, in forest fires, 2019 (39 fires); However, in the latter, the year 2010

It is followed in quantity (38), but in affected area, the most critical year has been 2016 (377.88 hectares) as shown in graph 25.

Graph 24 Forest fires reported per year in Bogotá D.C., between 1999 and September 2020



2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020

Burns Initiated Forest Fires

Source: District Environment Secretariat, based on the management reports of the District Commission for the Prevention and Mitigation of Forest Fires and the minutes of meeting 01, 2021 of that body.

Graph 25 Number of forest fires reported per year in Bogotá D.C., between 1999 and September 2020 and affected area

400
350
300
250 200 150
100
50 0

377,88
222,66
205,25
179,09
54,00
249,74
56,40
259,81
125,49
83,36
75,59
40,02
46,25
11,40 23,00
3,00
13,00
20,00
7,37 14,67
24,07
13,75

Quantity Affected area in hectares

Source: District Environment Secretariat, based on the management reports of the District Commission for the Prevention and Mitigation of Forest Fires and the minutes of meeting 01, 2021 of that body.